

Thermoelectricity I

(tutorial & inorganic TE materials)

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MATENER 2018

OVERVIEW

Basics and General Introduction.

Concepts. Figure of Merit.

Strategies to reduce thermal conductivity

Strategies to increase power factor

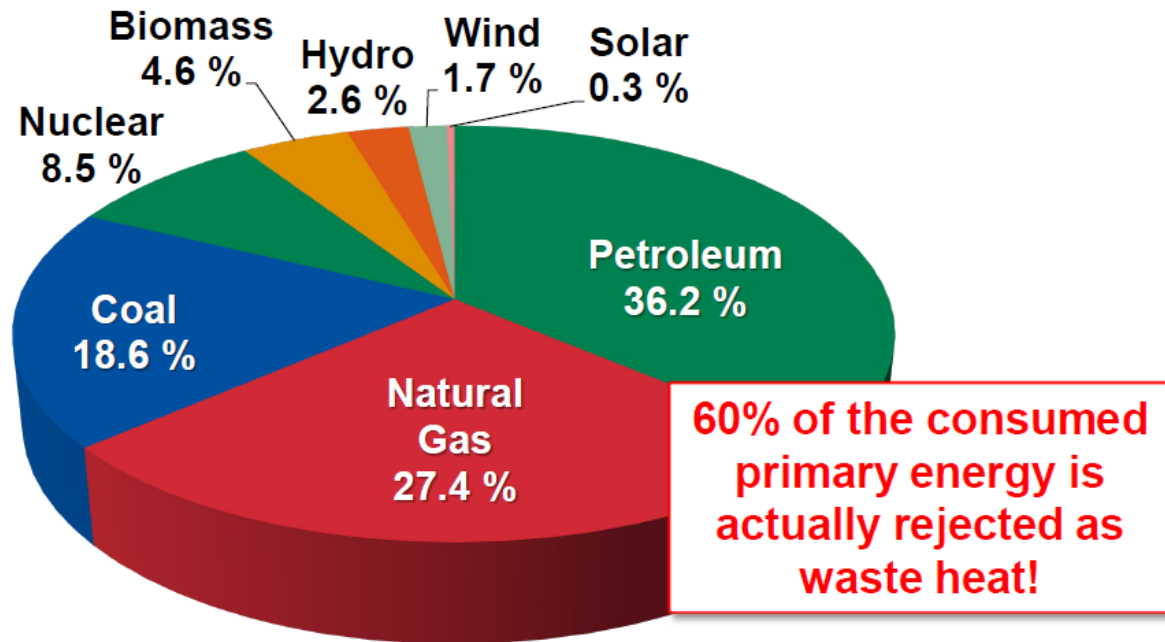
A short view on inorganic materials

Thermoelectric Devices & Applications

ENERGY SOURCES

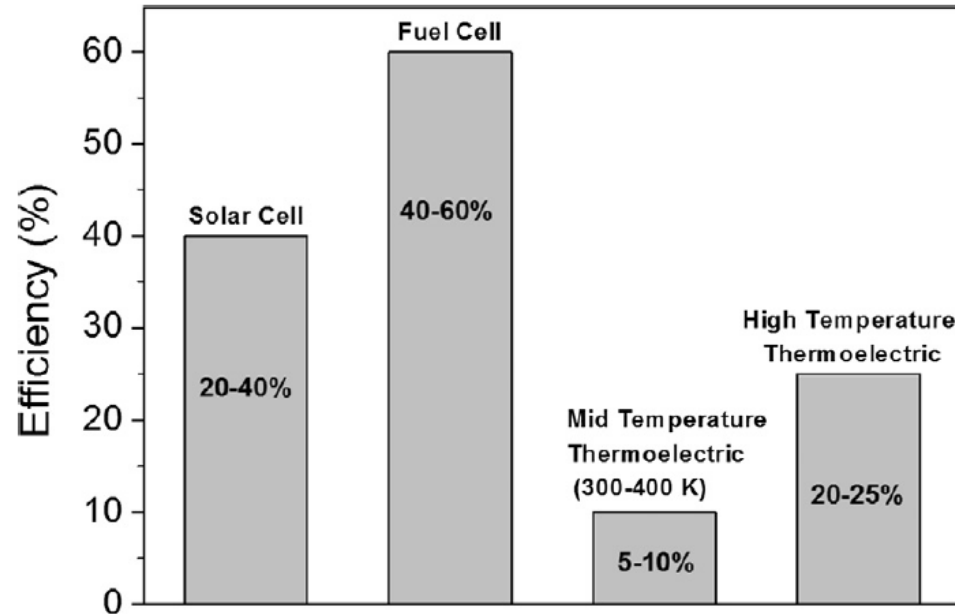
There is plenty of waste heat around

U.S. Energy Use in 2013



Convert waste heat into electrical energy using thermoelectrics?

Clean energy sources



Thermoelectric generation is still behind solar cell and fuel cell technologies

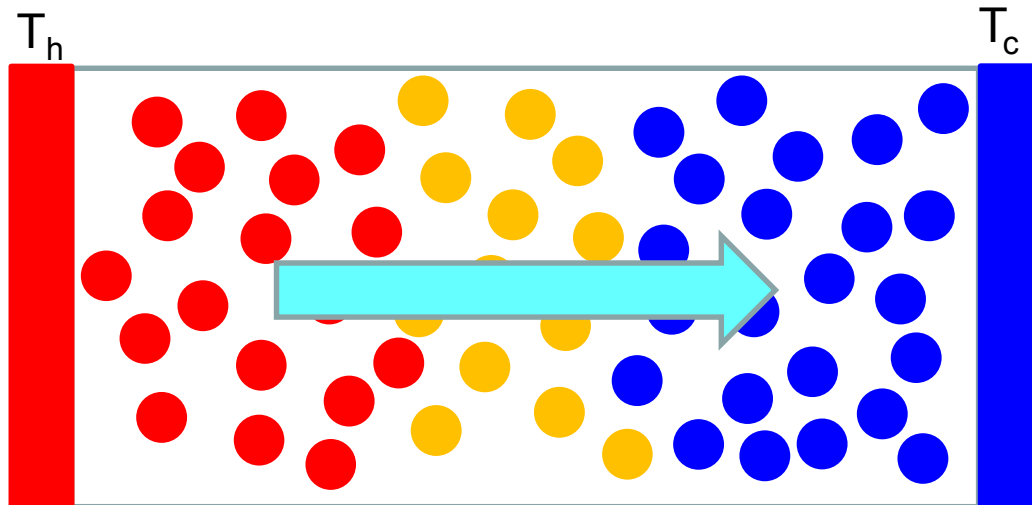
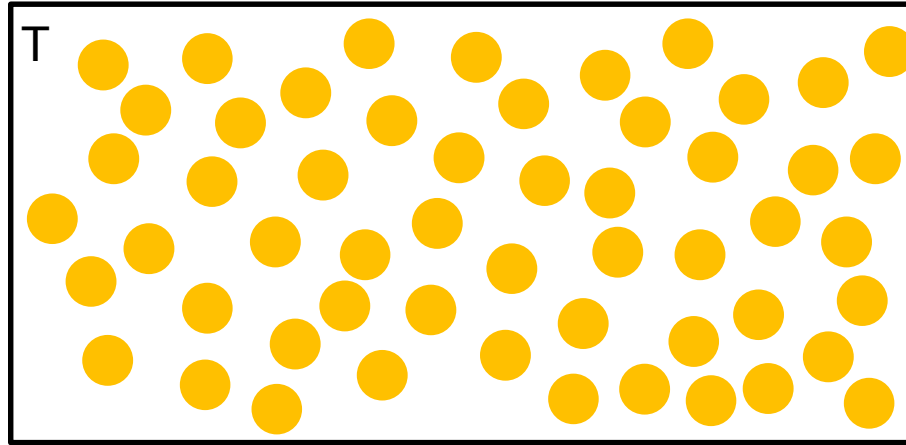
but

still useful to partially recover waste heat generated by combustion engines

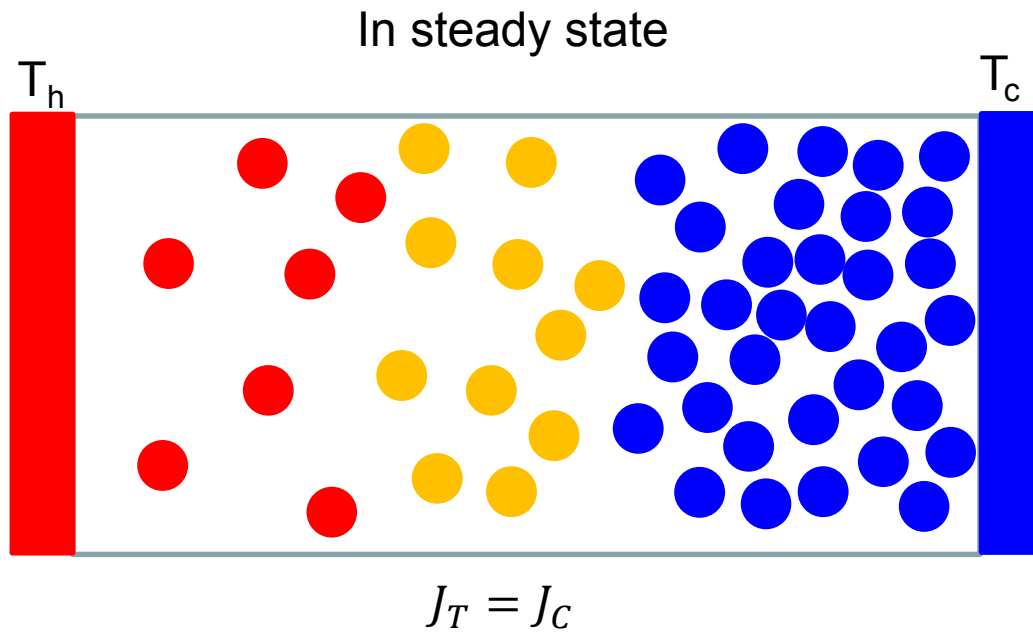
Principle

KINETIC THEORY

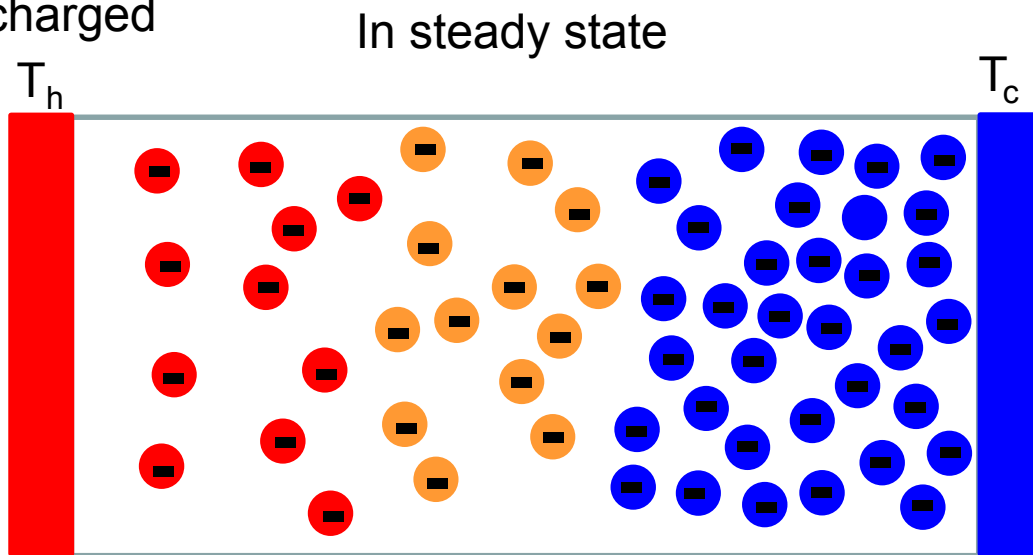
Gas particles
in equilibrium

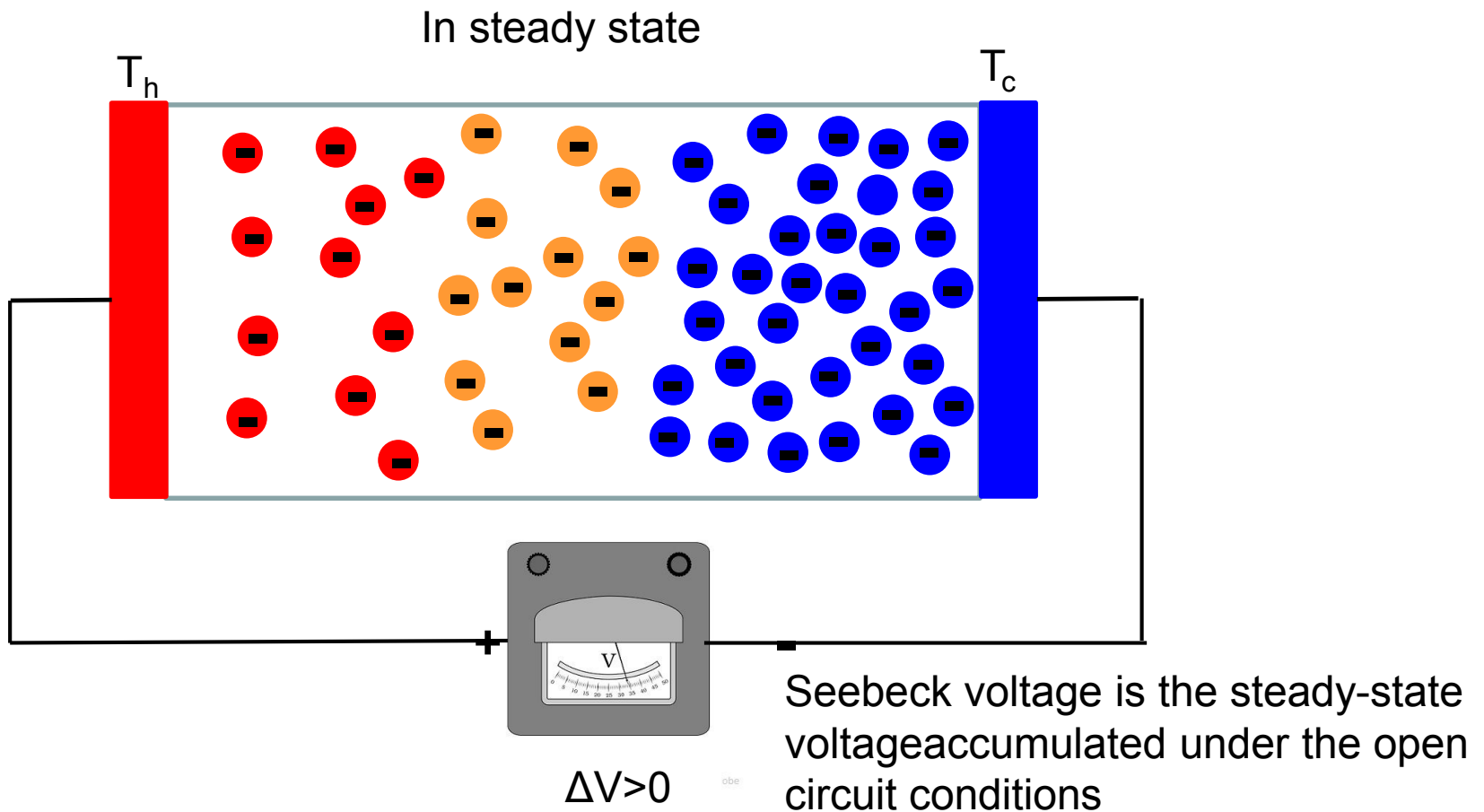


$$J_T \propto \nabla T$$



If particles are charged





Seebeck voltage (1821)

When a ∇T exists in bar a voltage develops between the two ends.

$$S = -\frac{V_h - V_c}{T_h - T_c} = -\frac{\Delta V}{T_h - T_c}$$

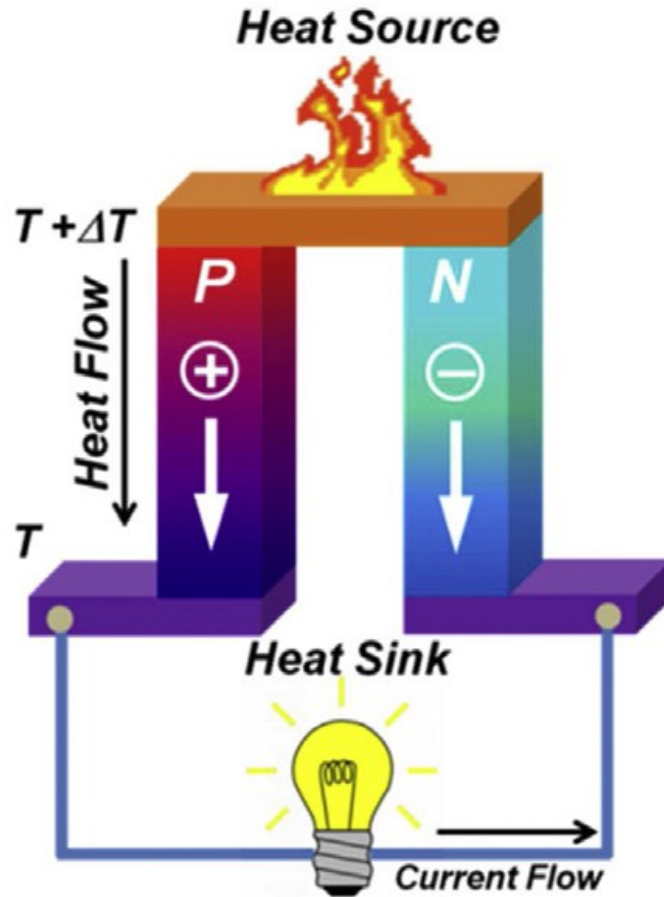
$S > 0$ p-type

$S < 0$ n-type

Except in very specific cases when slope is negative

$$\left\{ \frac{\partial \log \sigma(\epsilon)}{\partial E} \right\} < 0$$

Thermoelectric Generation



Power output

S thermopower

σ electrical conductivity

κ thermal conductivity

The idoneity of a TE material can be evaluated by

$$ZT = \frac{S^2 \sigma}{\kappa} T$$

FIGURE OF MERIT

FIGURE of MERIT & EFFICIENCY

Carnot efficiency

$$\eta_C = \frac{T_h - T_c}{T_c}$$

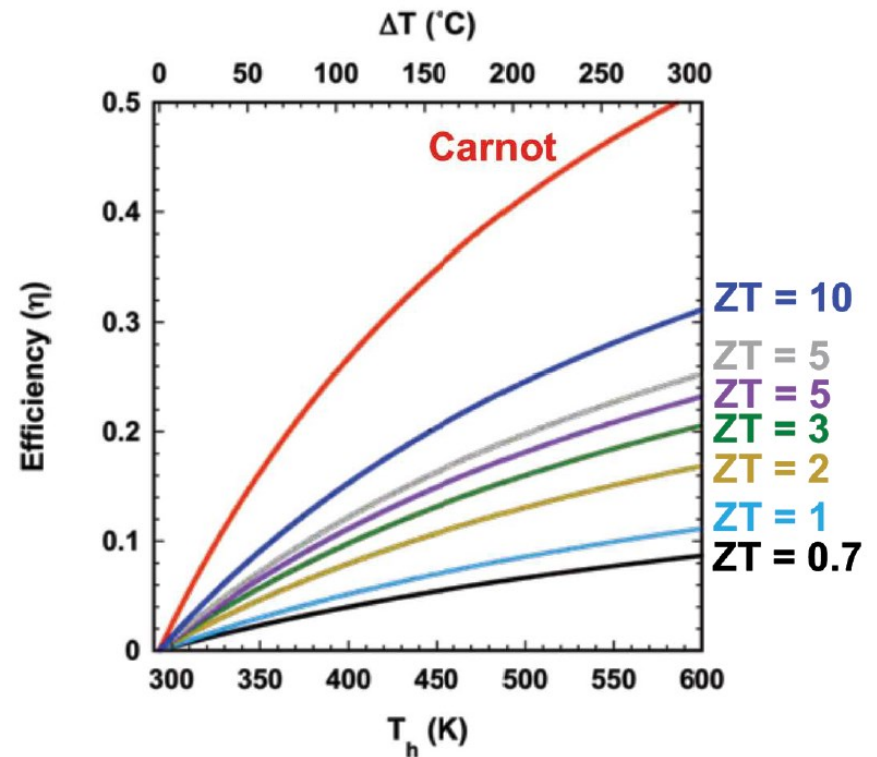
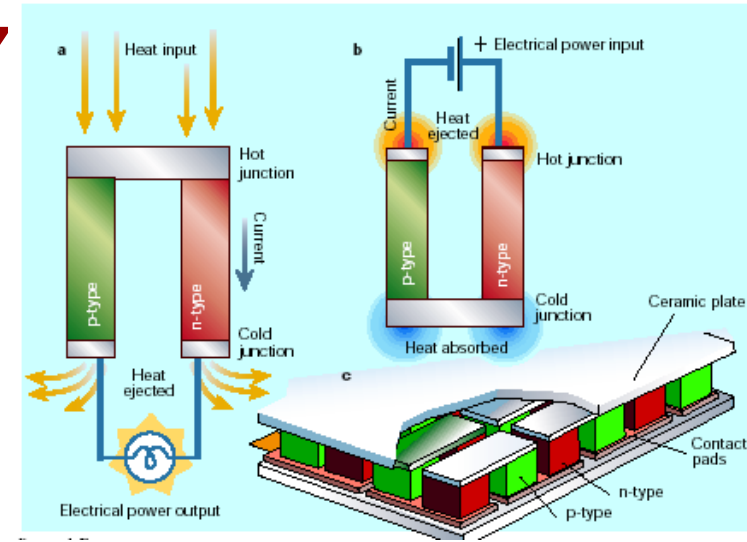
Thermoelectric efficiency

$$\eta_{TE} = \eta_C \cdot \frac{\sqrt{1 + ZT} - 1}{\sqrt{1 + ZT} + \frac{T_c}{T_h}}$$

$ZT < 1 \rightarrow$ very specialized applications

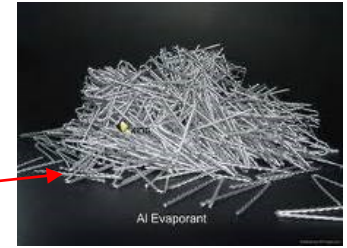
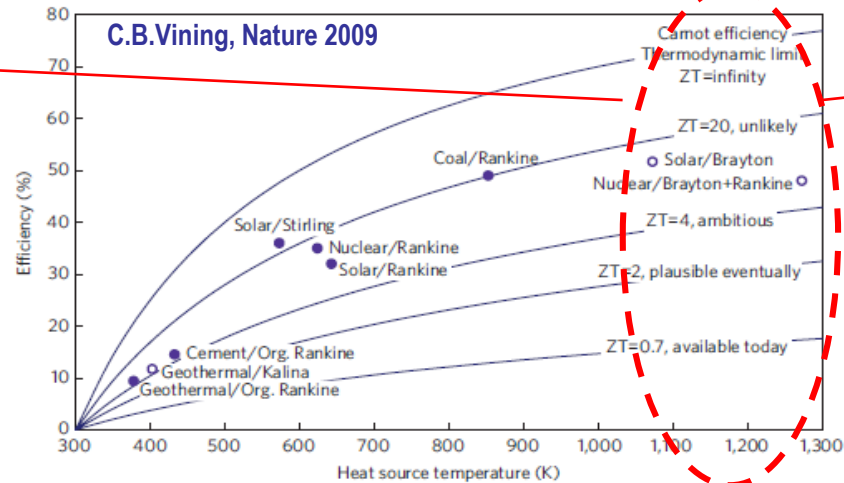
$ZT > 1 \rightarrow$ small-scale applications

$ZT > 3 \rightarrow$ large-scale applications (currently, $ZT < 2$)



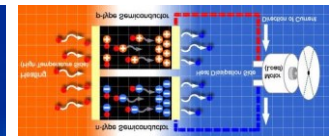
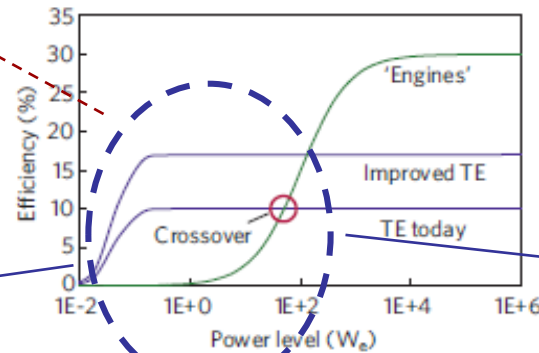
Impact of ZT on power applications

Large scale



Medium

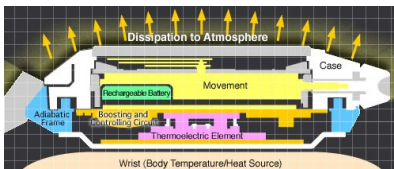
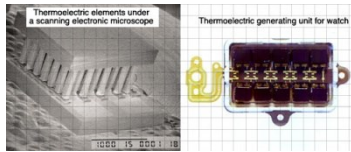
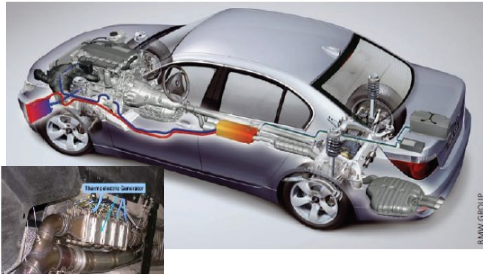
Vining, Nature 2009



Small scale

Advantages: No moving parts, little maintenance, no greenhouse gas emission

High-T

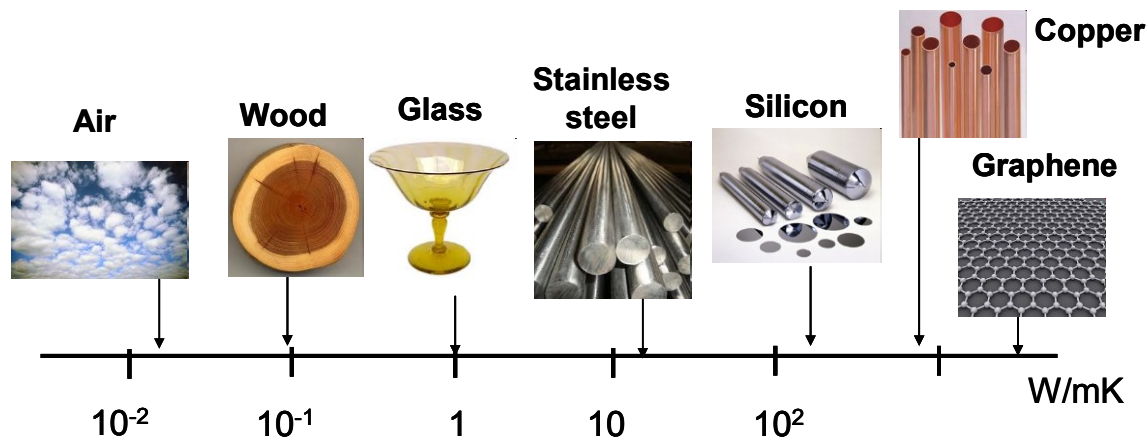


Low-T

High efficiency requires maximizing ZT $\rightarrow$ Increase in $S^2\sigma$
 Decrease in κ $\rightarrow$ PHONON GLASS
 ELECTRON CONDUCTOR (Slack, 1994)

$$ZT = \frac{S^2\sigma}{k}T$$

No such material exists in Nature !!!



**Good heat conductors
are
good electrical conductors**

$$k = k_l + k_e$$

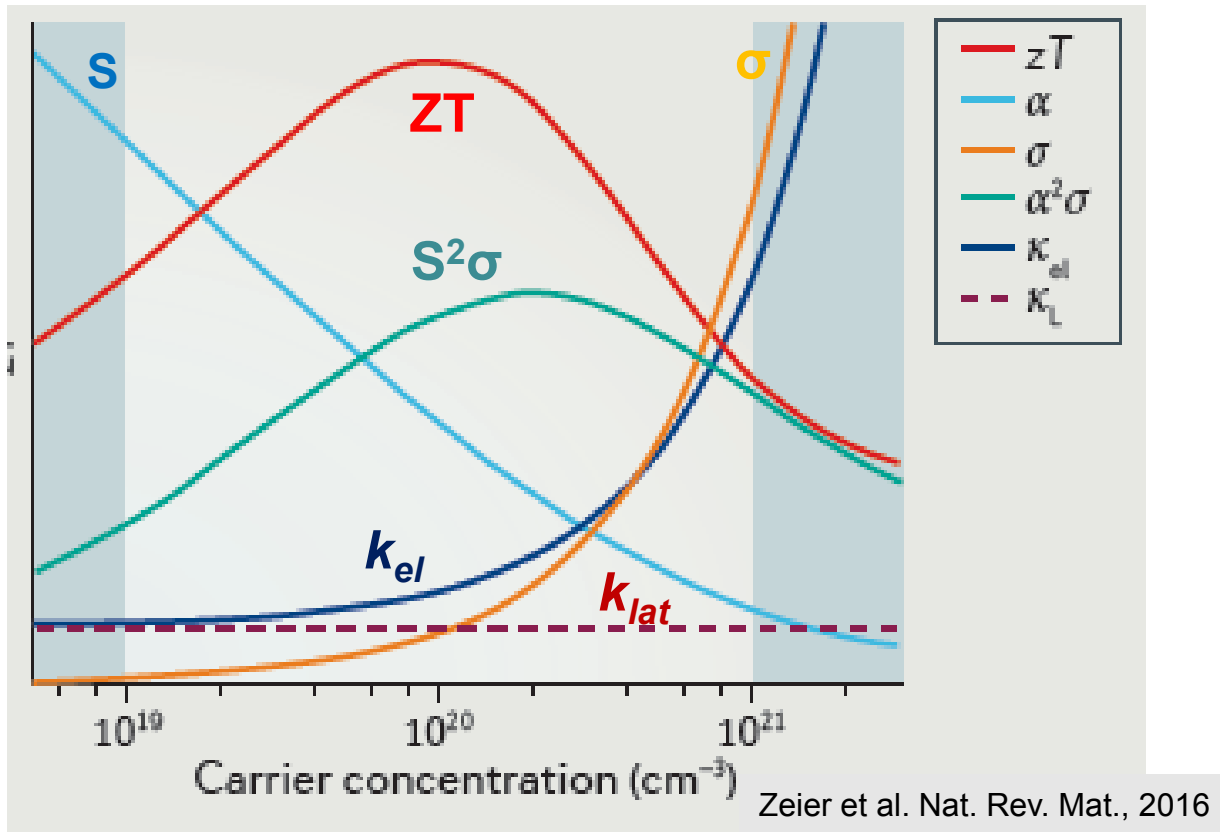
metals have lots of free electrons

(k_e is large), while ceramics

have few (only k_l is active)

Material	k (W/m-K)	Energy Transfer
• Metals		
Aluminum	247	By vibration of atoms and motion of electrons
Steel	52	
Tungsten	178	
Gold	315	
• Ceramics		
Magnesia (MgO)	38	By vibration of atoms
Alumina (Al ₂ O ₃)	39	
Soda-lime glass	1.7	
Silica (cryst. SiO ₂)	1.4	
• Polymers		
Polypropylene	0.12	By vibration/rotation of chain molecules
Polyethylene	0.46-0.50	
Polystyrene	0.13	
Teflon	0.25	

Tuning Seebeck, thermal & electrical conductivity



Large values of ZT require high S , high σ and low κ $\longrightarrow$ Difficult to accomplish

$S \uparrow \longrightarrow \sigma \uparrow$

$\sigma \uparrow \longrightarrow \kappa \uparrow \longrightarrow$ Wiedemann-Franz law

Strategies to reduce thermal conductivity

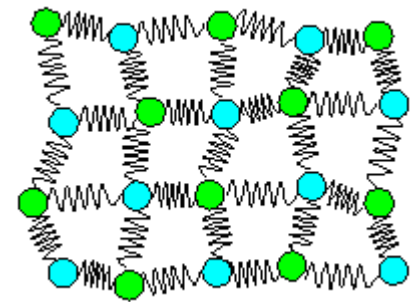
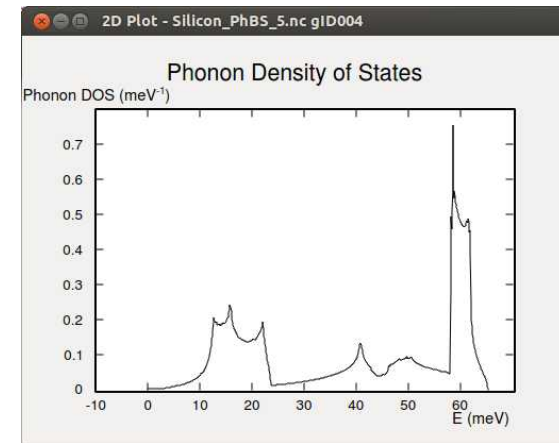
Phonons (heat carriers)

Quantized energy of lattice vibration

$$E = \hbar\omega = h \frac{c}{\lambda}$$

Based on spring and mass system

Phonons scattered whenever there
is a difference
in mass and/or spring constant.



Heavy mass-concept

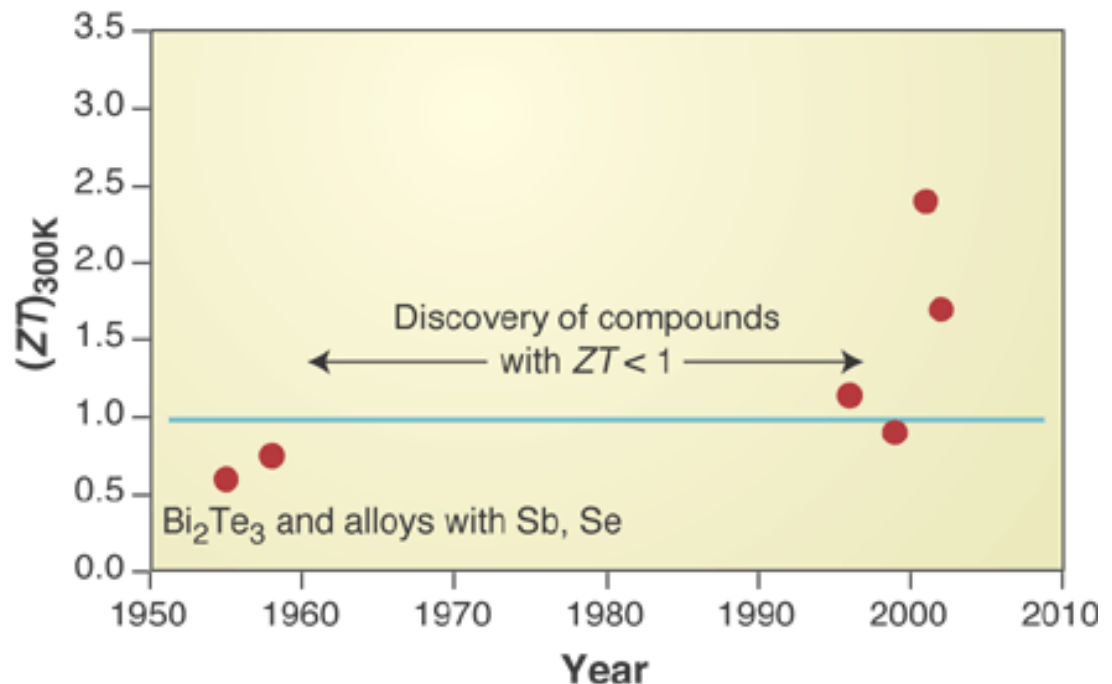
1950–1959: Ioffe stated that the thermal conductivity of semiconductors is a function of atomic weight. Generally, elements with large atomic weights have low thermal conductivity.

$$k = \frac{1}{3} C \cdot v \cdot l$$

Reducing κ without affecting S and σ is realized in semiconductors of high atomic weight which reduces the speed of sound in the material and therefore κ .

STRATEGY USED 1950-1990

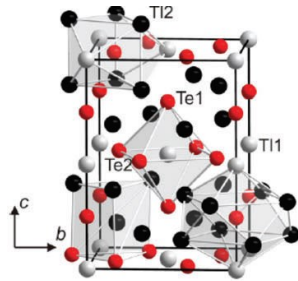
BiTe
PbTe
SbTe
and alloys



Engineering phonon thermal transport

Reducing k without as much reduction of σ and S

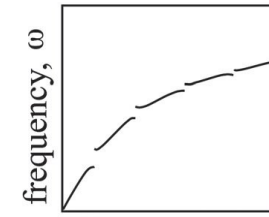
Engineering phonon thermal transport



Large unit cell

$$\lim_{q_{ph} \rightarrow 0} v = c_s \sim \sqrt{\frac{E}{m}}$$

Heavy mass, m & low elastic modulus, E

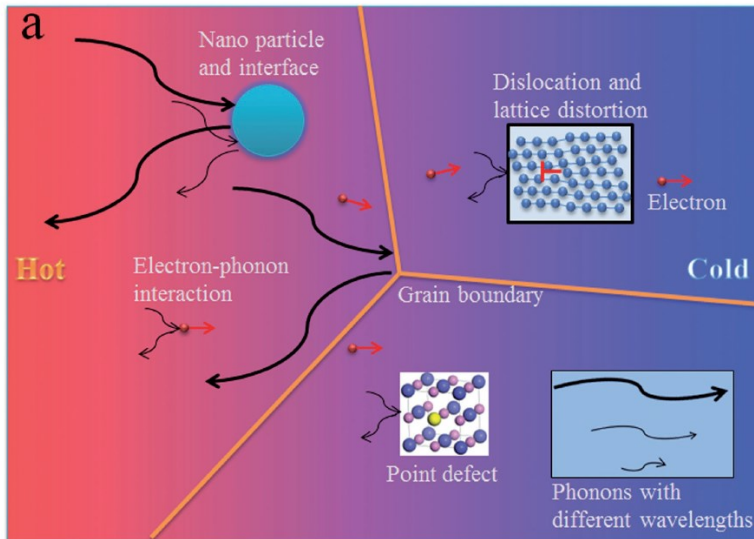
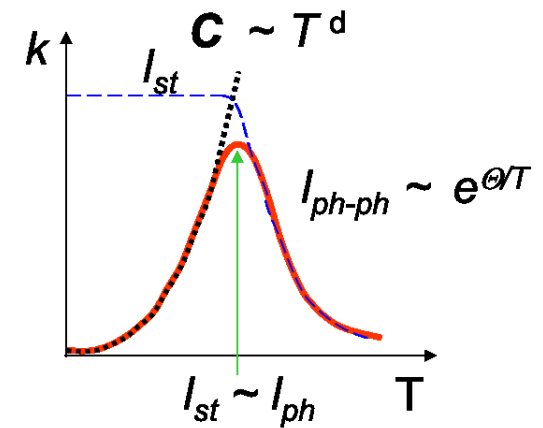


wavevector, q_{ph}

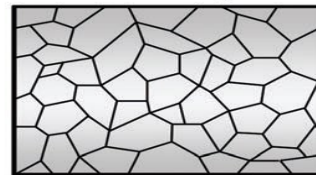
Phonon bandgaps

$$\kappa_{lat} = \frac{1}{3} \cdot \int C(\omega) \cdot l(\omega) \cdot v(\omega) \cdot d\omega$$

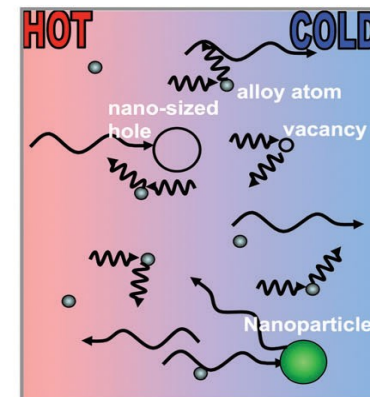
scattering



Grain-boundary scattering

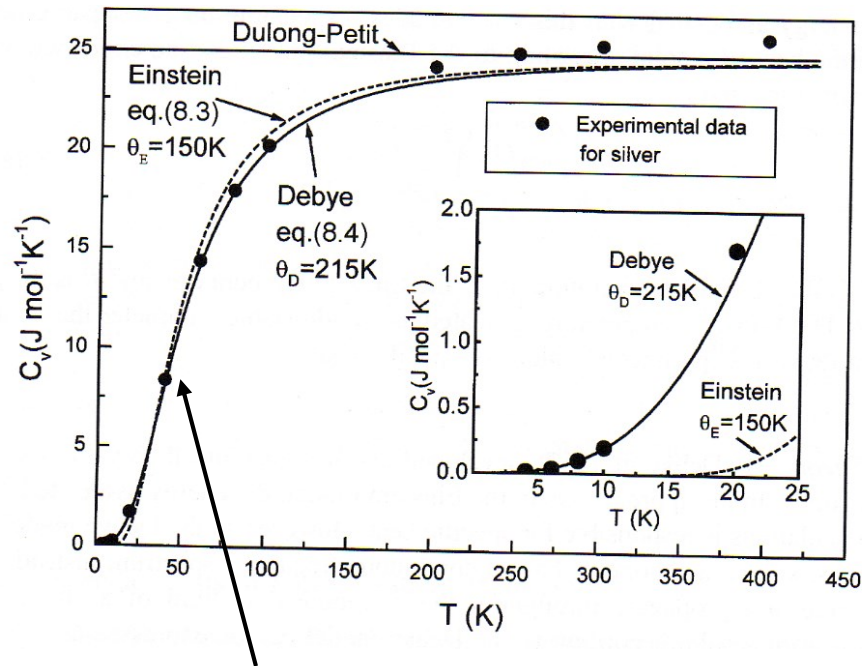


Impurity scattering

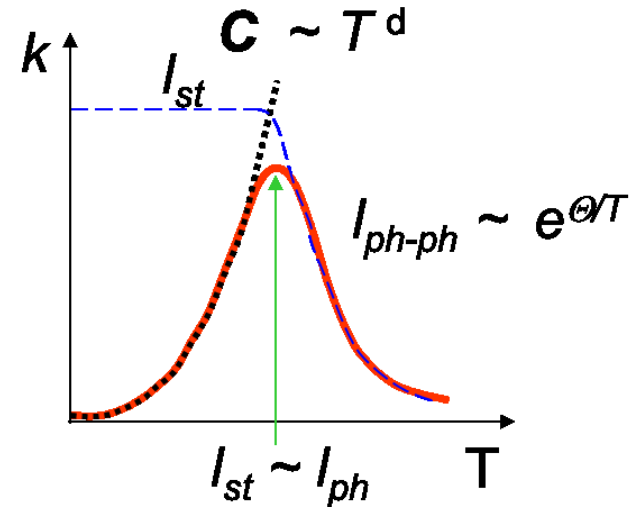


Specific heat $\longrightarrow C_V = \left(\frac{\partial E_{ph}}{\partial T} \right)_V$

If $T > \Theta$, $C \sim \text{constant}$



Debye-like regime T^3



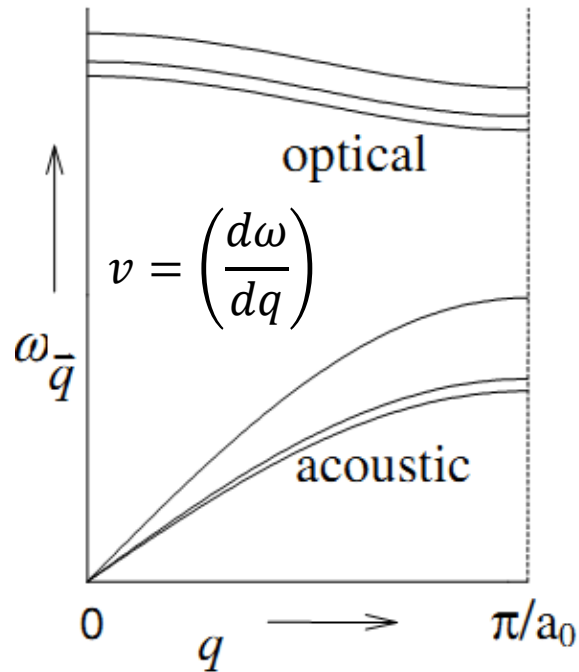
If $T \ll \Theta$, $C \sim T^d$ (d: dimension)

Size-effects: useful at very low-T's

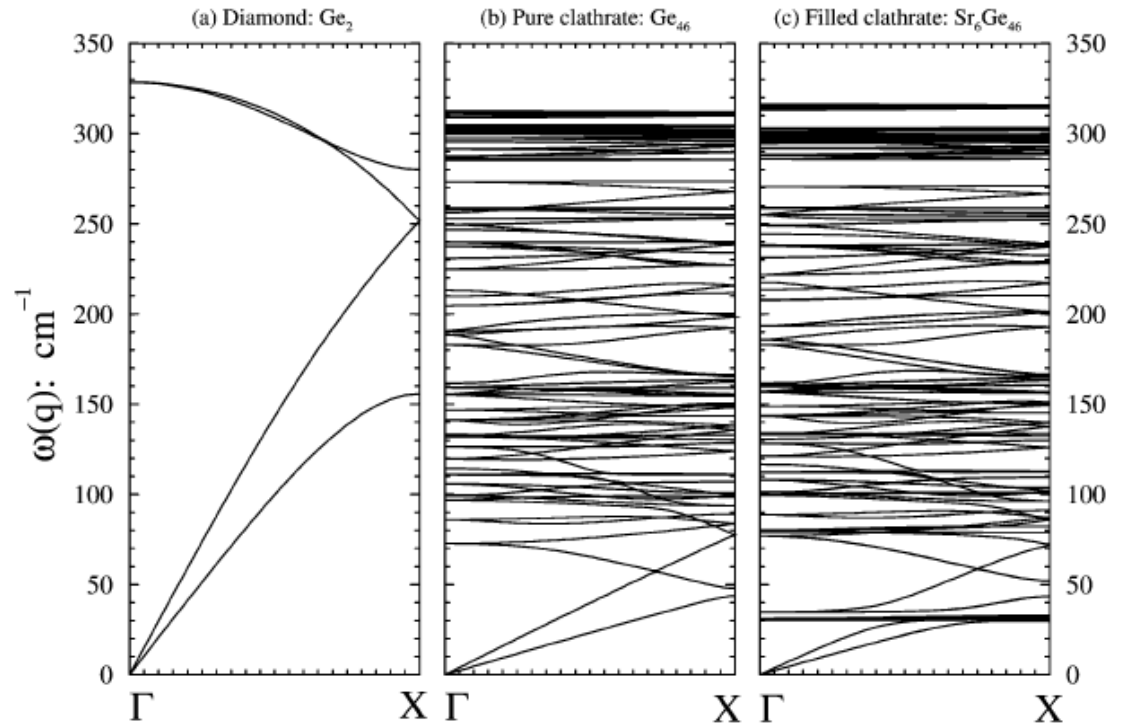
Size of the crystal is of atomic scale so lattice vibrations are confined and dispersion relation is modified

Phonon group velocity

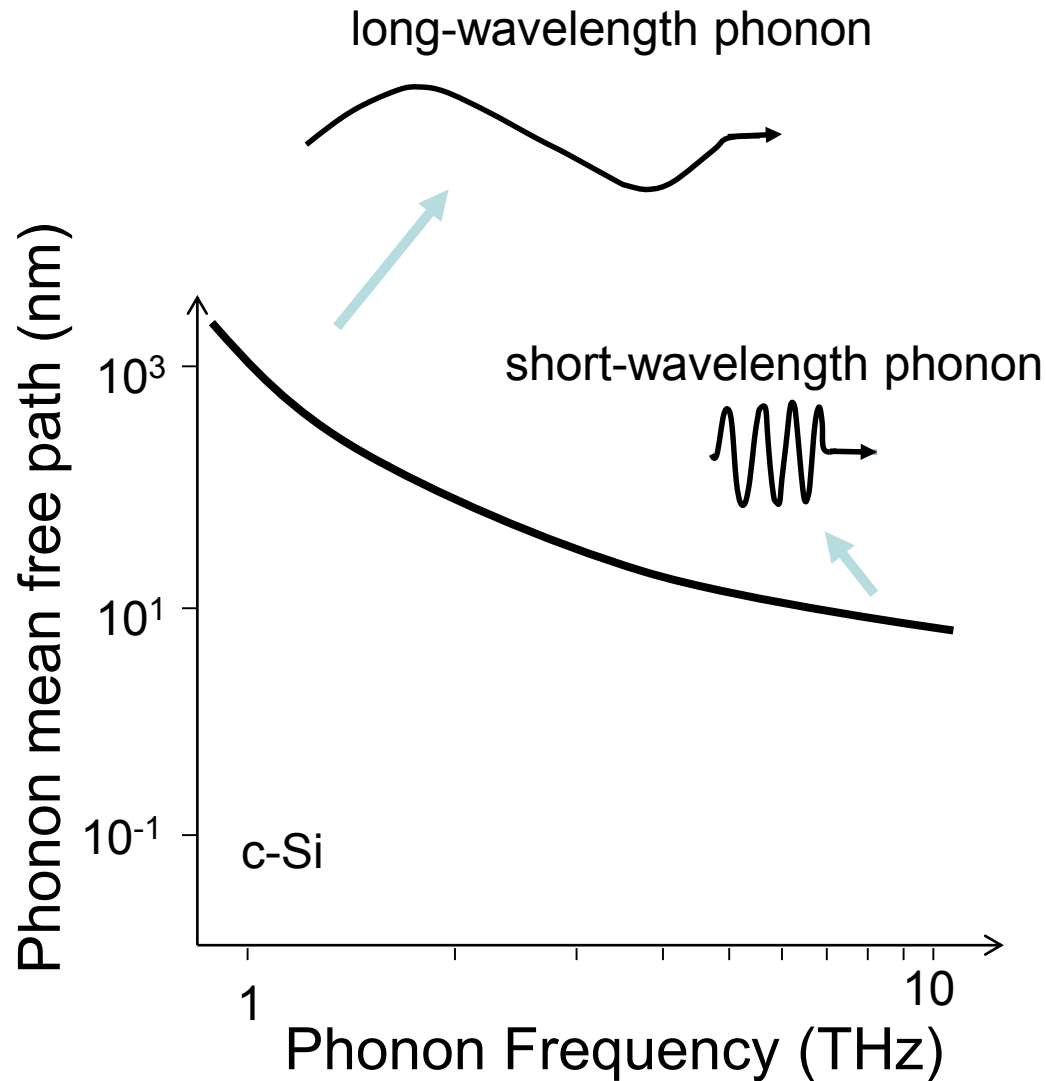
Optical branches:
Small sound velocity



i) Increasing the number of atoms in the unit cell



Phonon mean free path (average distance between collisions)



At room T

PHONONS

Wavelength	Mean freepath
------------	---------------

Dominant heat-carrying 5-10 nm but broadband	10-100's nm
--	-------------

ELECTRONS

Wavelength	Mean freepath
------------	---------------

1-3 nm	1-5 nm
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$$\frac{1}{l_{eff}} = \frac{1}{l_b} + \frac{1}{l_i} + \frac{1}{l_{ph}}$$

l_b boundary scattering
 l_i impurity scattering
 l_{ph} phonon-phonon scattering

Wien's law holds for phonons

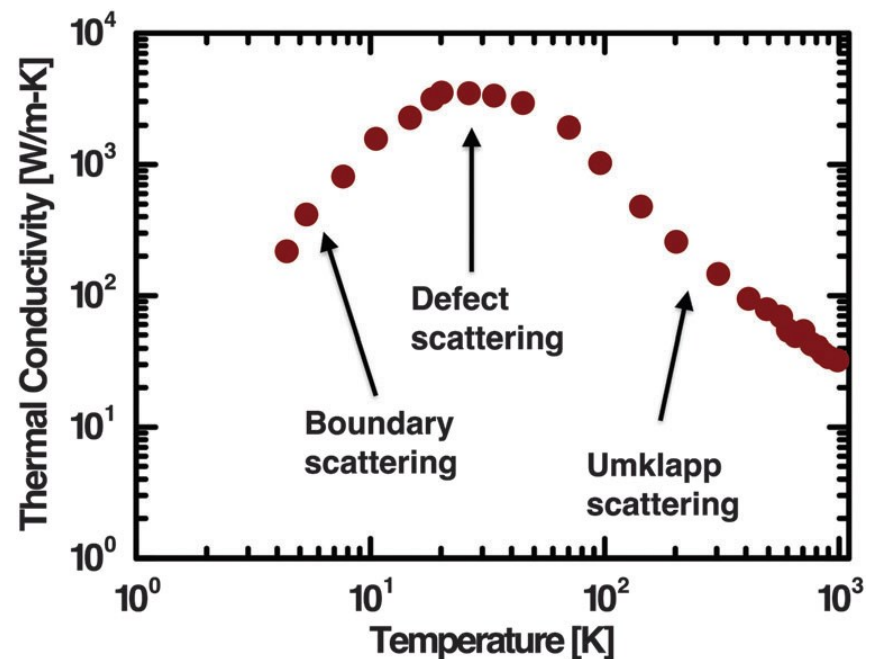
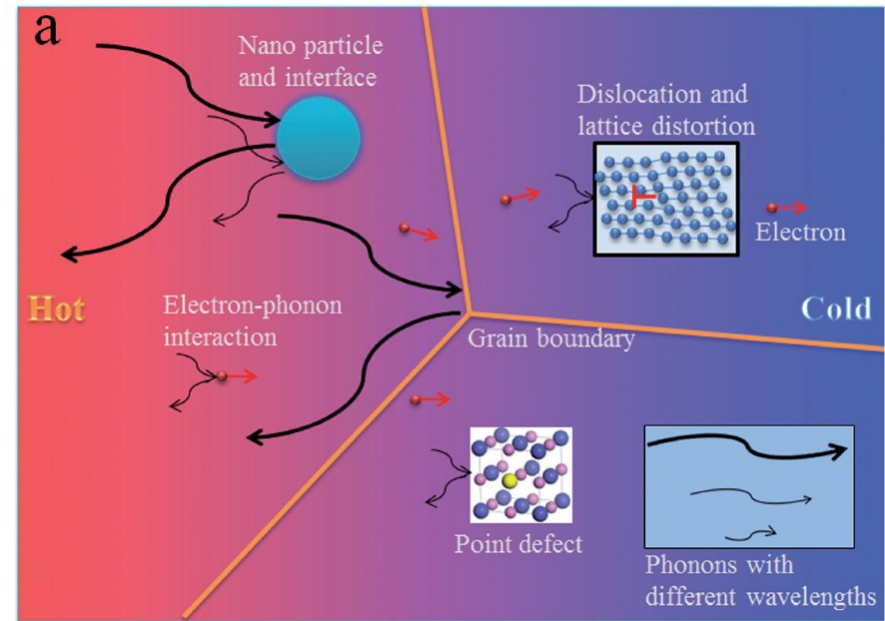
$$\lambda_{max} \cdot T \approx cte$$

Long λ phonons carry most heat at low T

Boundary scatt. at low-T

As T increases λ_{ph} dominant phonons decreases

At high T ph-ph scattering

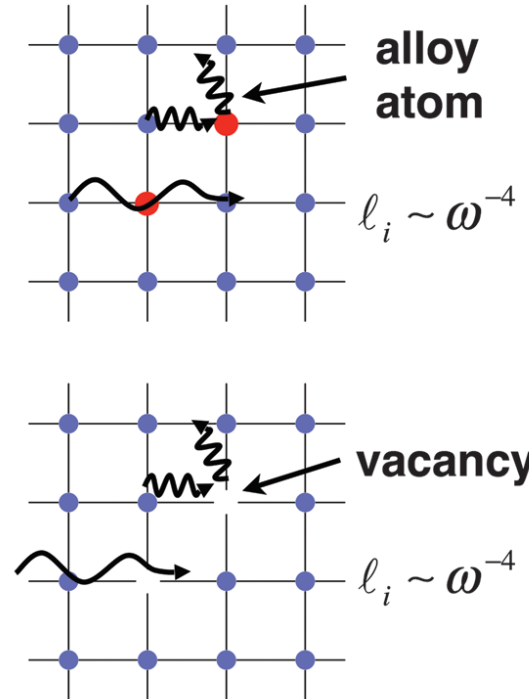


DECREASING THE MEAN FREE PATH OF PHONONS: TOWARDS LOW THERMAL CONDUCTIVITY

ALLOY LIMIT

Alloys: Lowest thermal conductivity in crystalline solids

Atomic substitutions heavily scatter phonons



Rayleigh scattering regime: scattering cross section

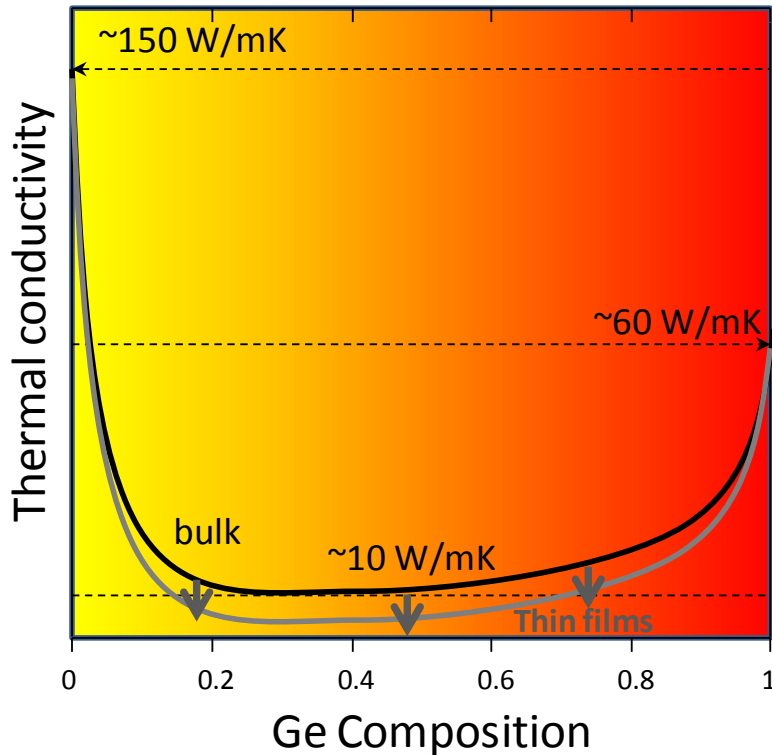
$$\sigma \sim b^6/\lambda^4$$

If atoms have $b \sim 1\text{\AA}$ $\Rightarrow$ Alloys scatter short-wave phonons more effectively than mid-long-wave phonons

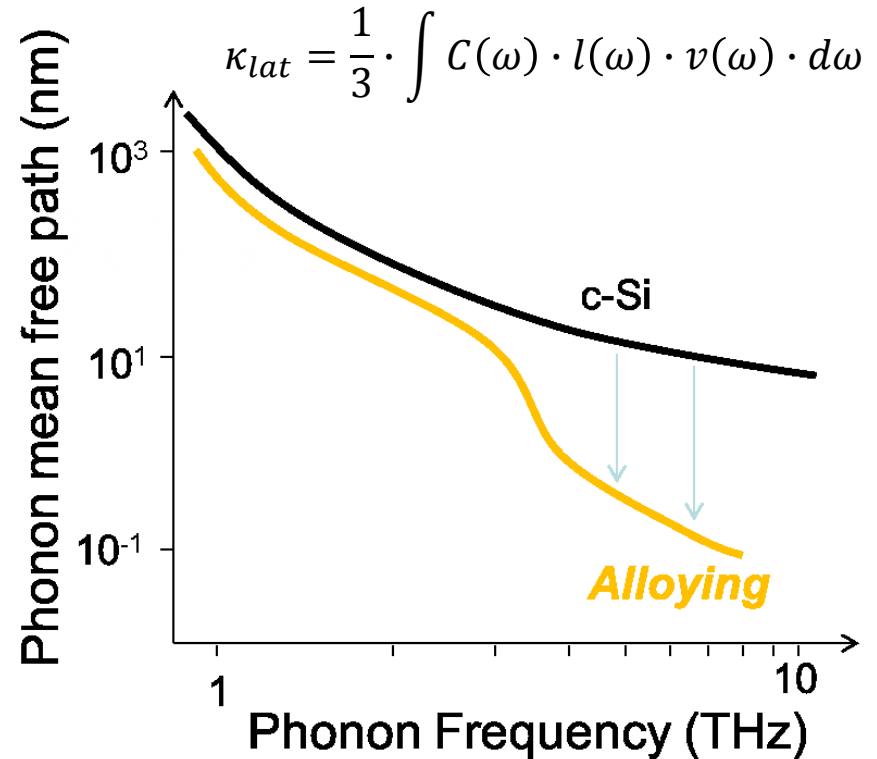
ALLOYING



Mainly affects short-wavelength
(high frequency) phonons



Mass disorder induces
strong reduction
of thermal conductivity



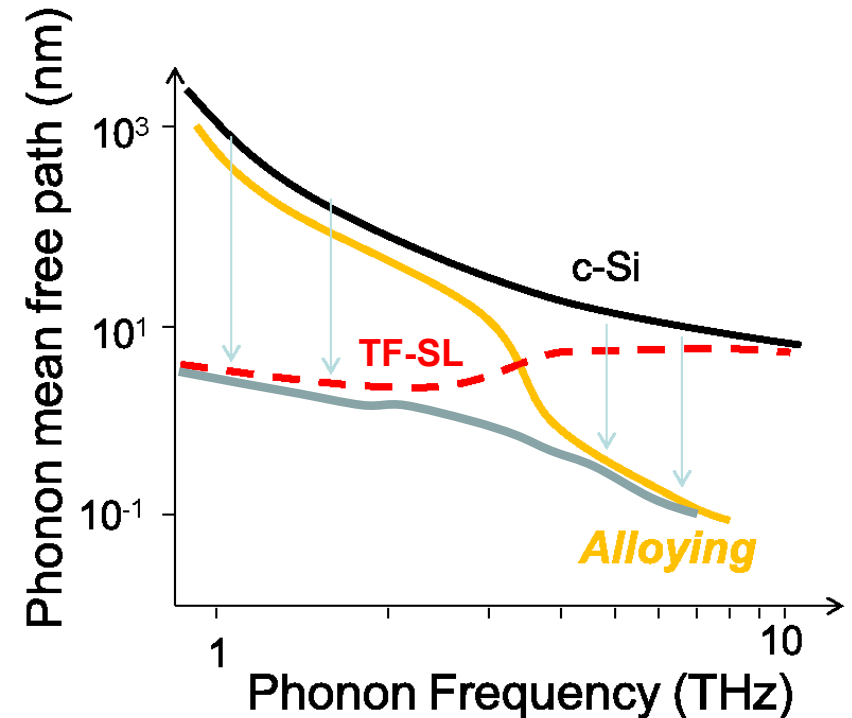
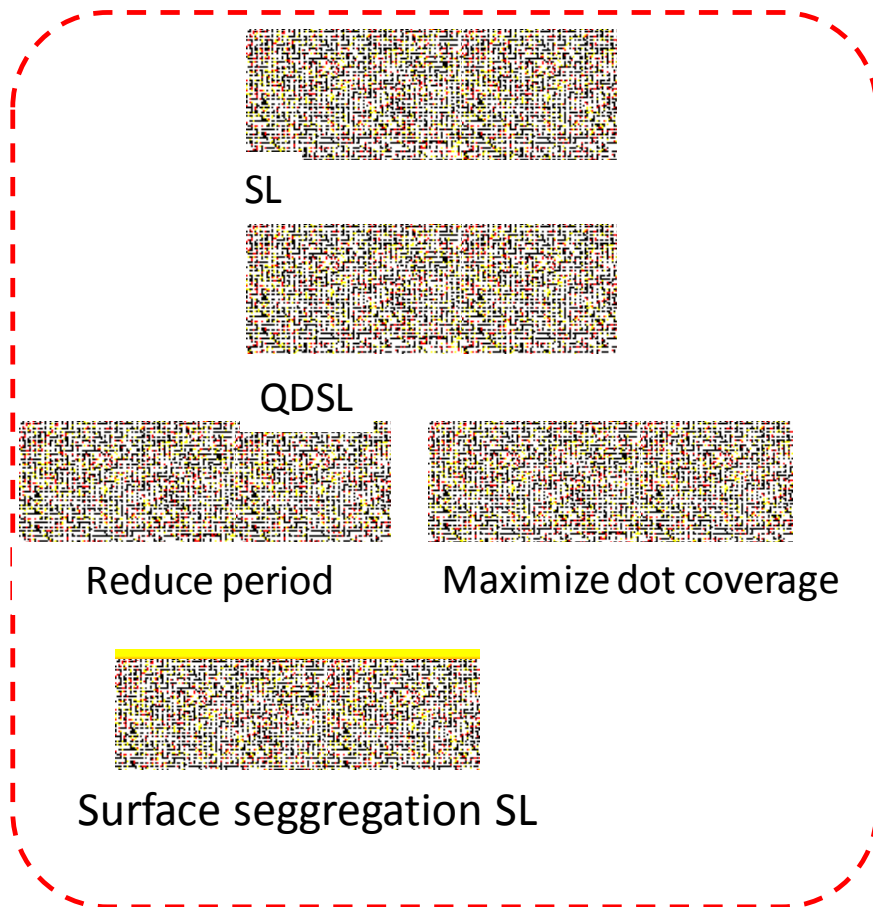
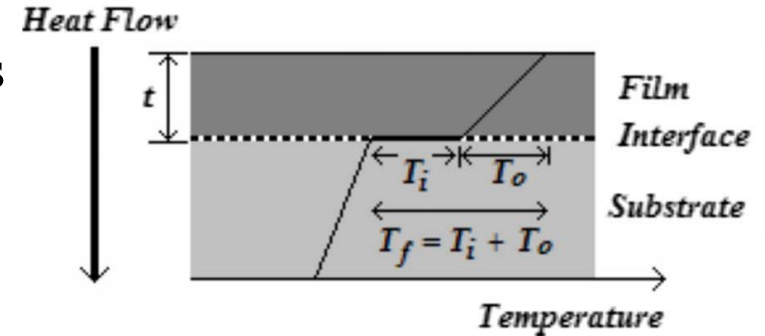
Maximum variation below 10 at % Ge
 k constant between 20-70 at % Ge

If mid- to long-phonons could be scatterered, then alloy limit can be beaten !!!!!

THIN FILMS & SUPERLATTICES

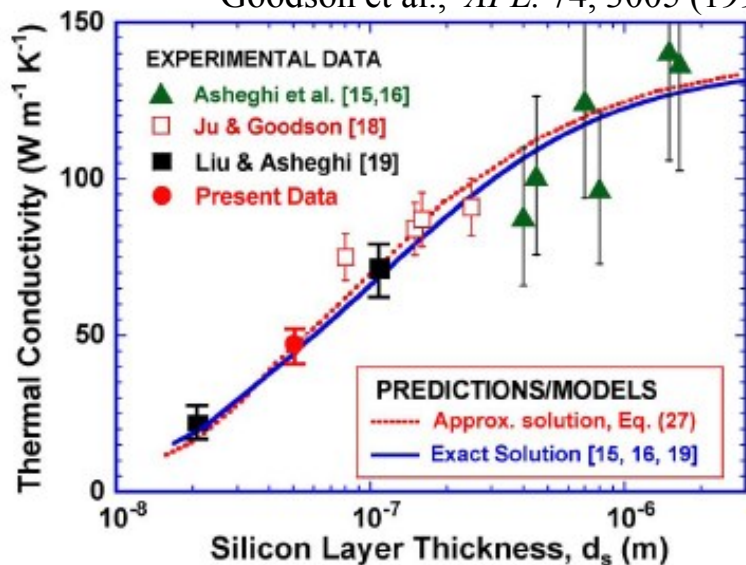
Phonon-blocking/electron transmitting interfaces in Si/Ge and other SLs

Use of the acoustic mismatch between the SL components to reduce κ_{ph} .

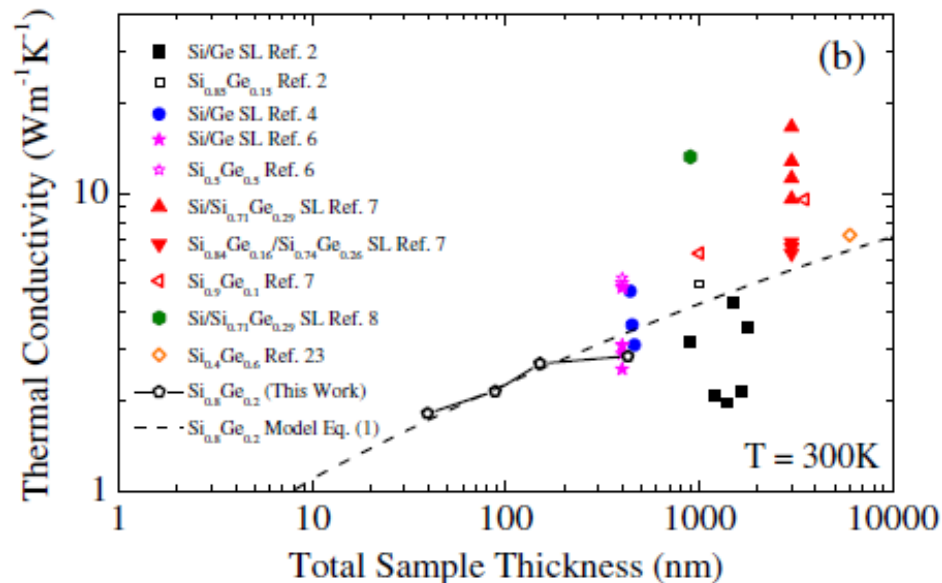


Thin films of Si

Goodson et al., *APL*. 74, 3005 (1999)

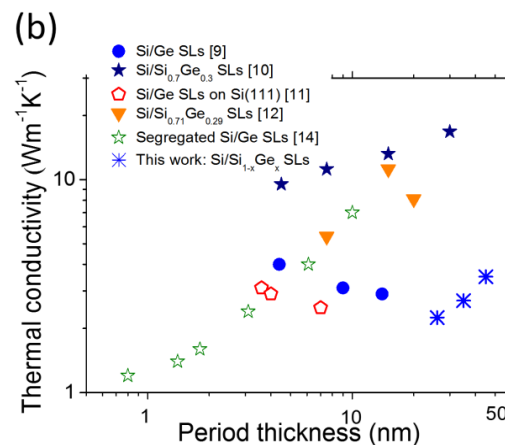
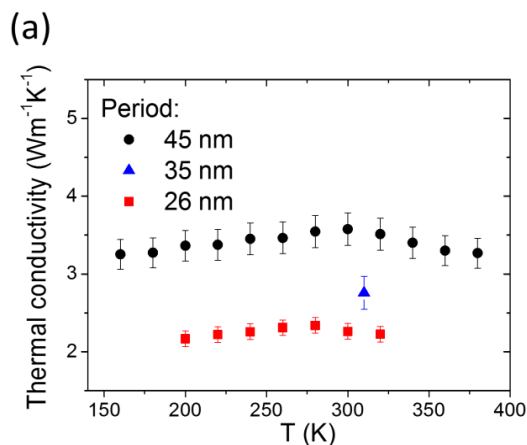
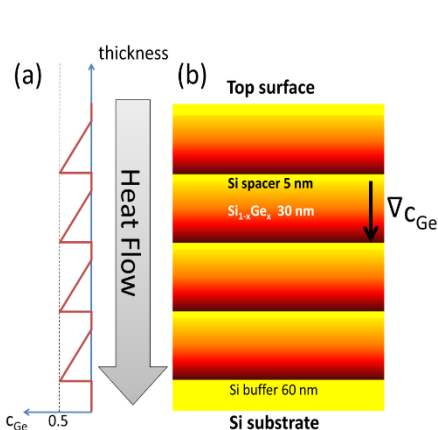


Thin films of SiGe alloys

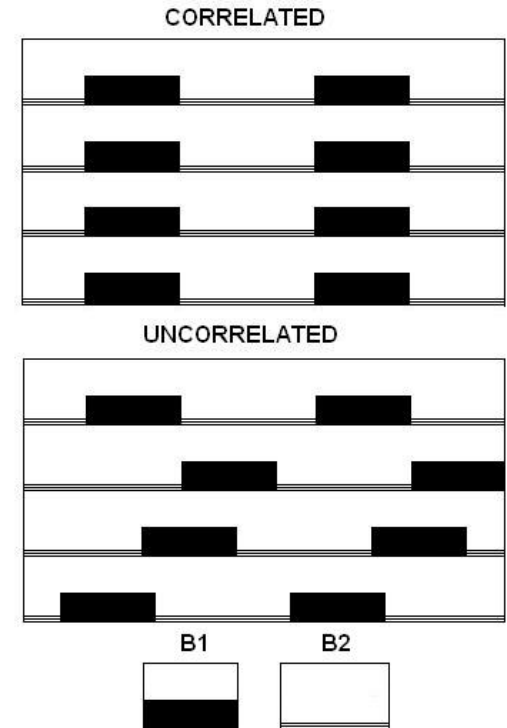
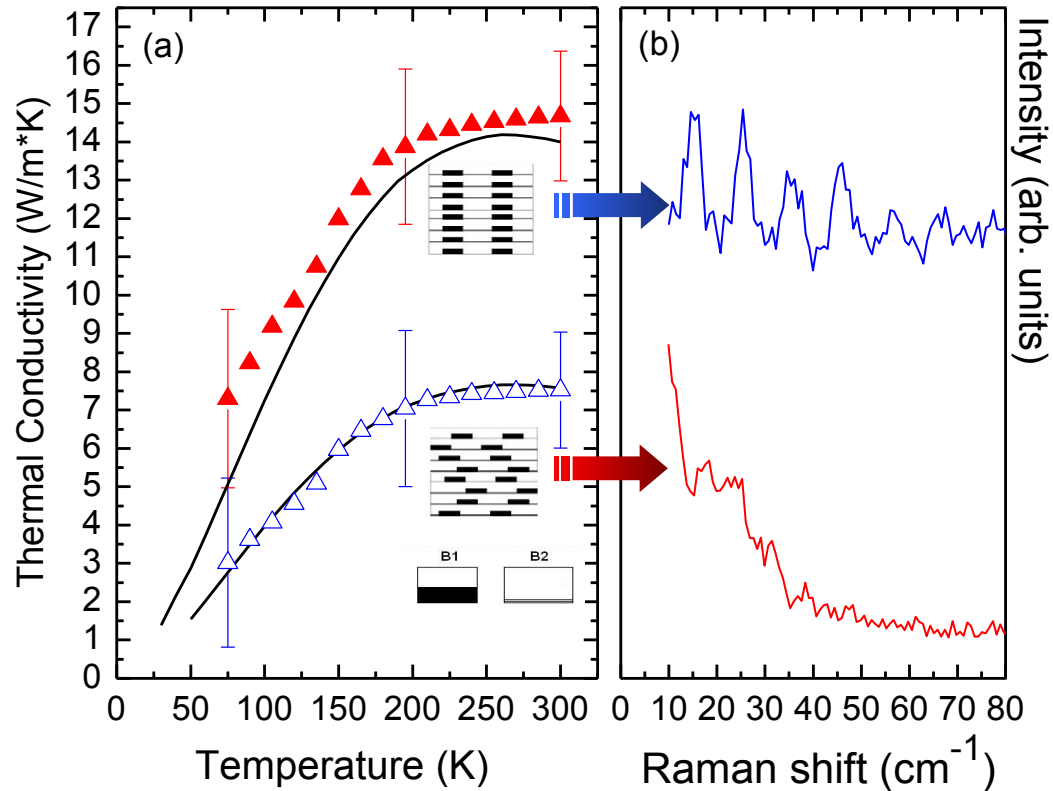


Cheaito et al. PRL 2012

SiGe Superlattices



Thermal conductivity of Ge Quantum Dot Superlattices

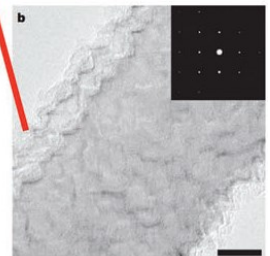
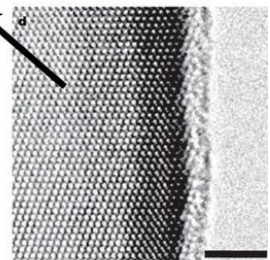
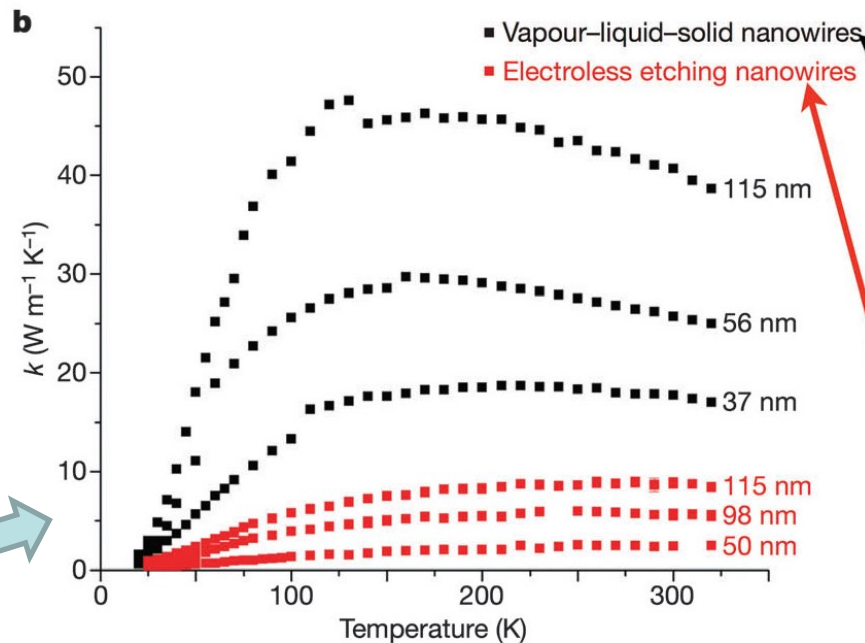
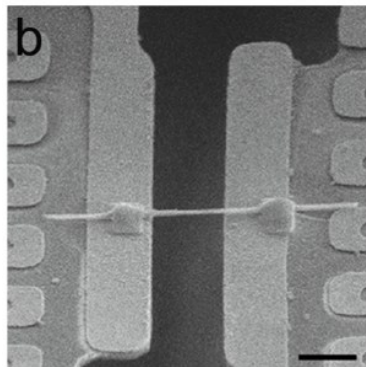
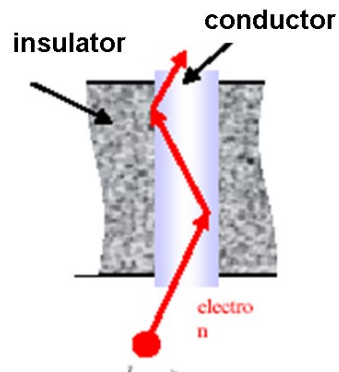


Lower-D MATERIALS (nanowires)

Increase boundary scattering of phonons at the barrier-well interfaces, without as large increase in electron scattering at the interface

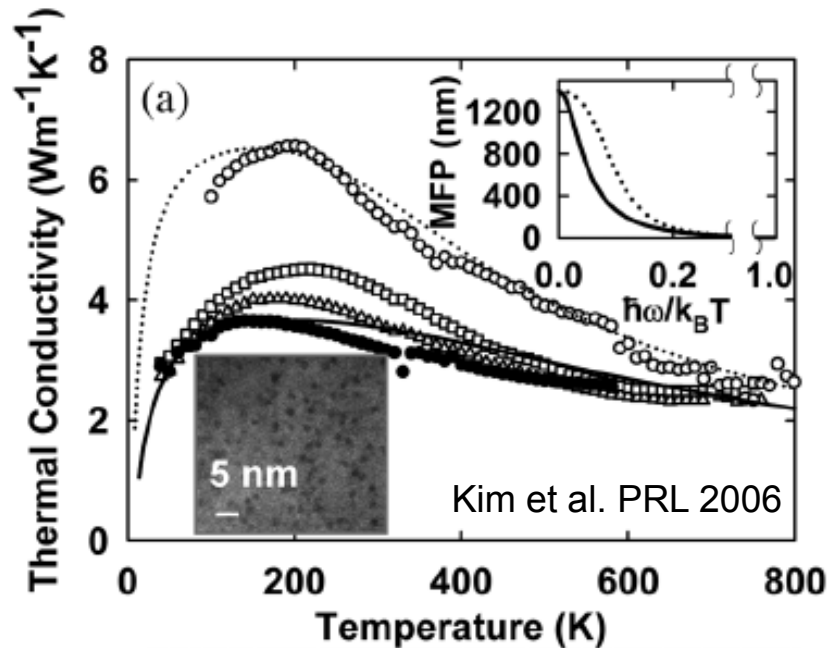
$\kappa \downarrow$ while keeping $\sigma \uparrow$

If the width of the semiconductor is smaller than the mean free path of phonons and larger than that of electrons or holes

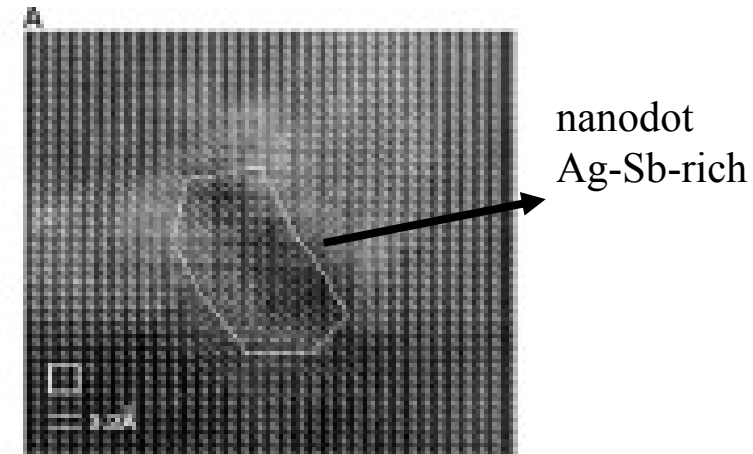


Embedded nanostructures in semiconductor alloys

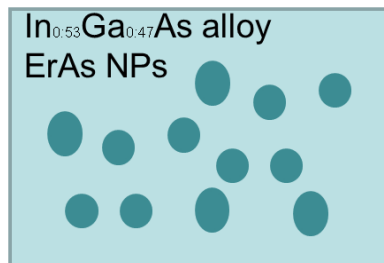
scatter the mid-to-long wavelength phonons more efficiently because the similarity in sizes, reducing k below the alloy limit.



$\text{AgPb}_{18}\text{SbTe}_{20}$ Hsu et al. Science 303 (2004)



Reduction by 2 below alloy limit



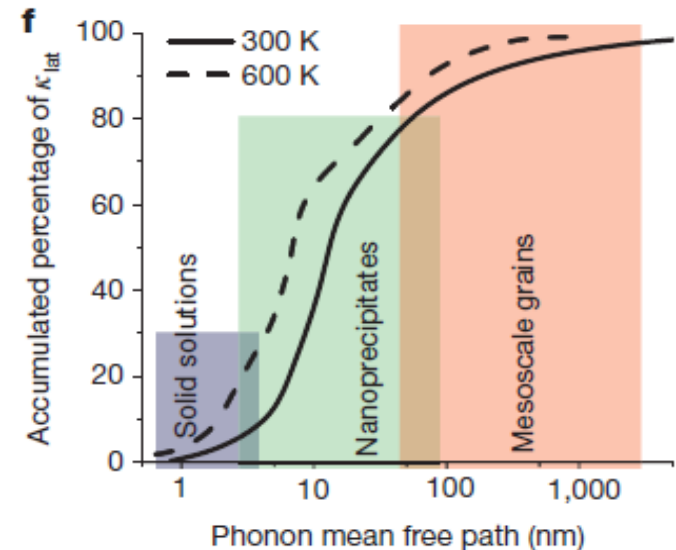
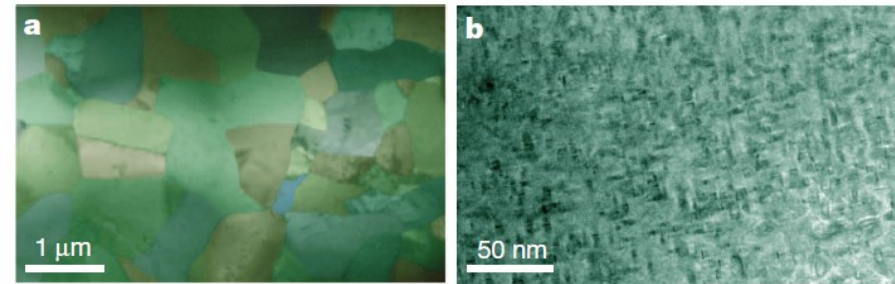
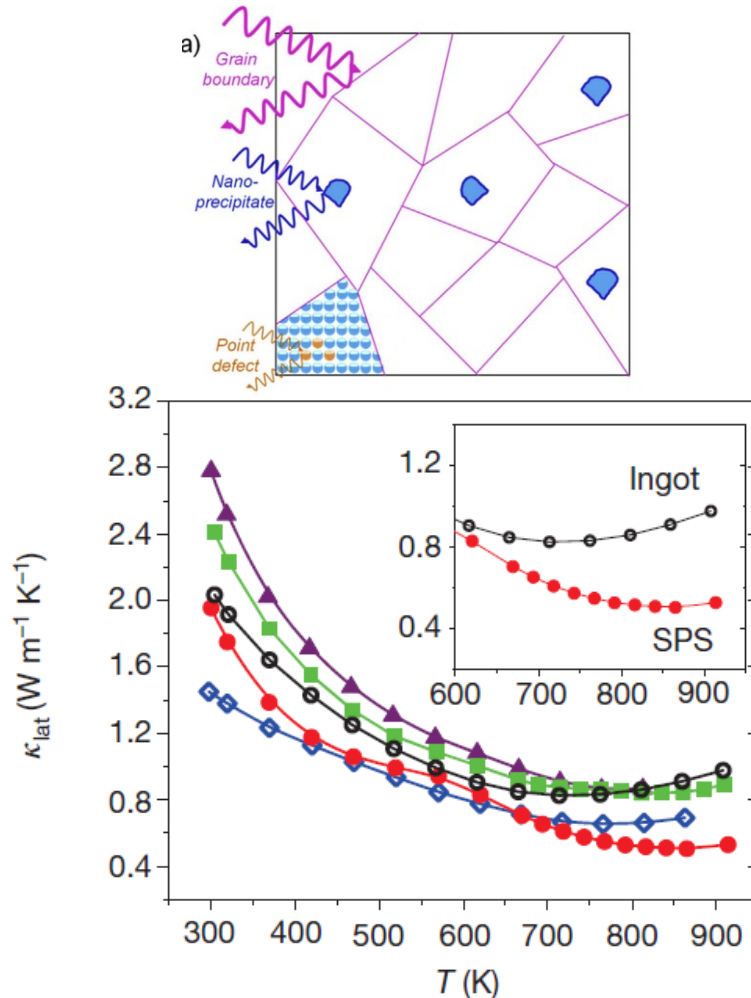
$$k_{\text{lat}} = 1.30 \text{ W/mK at } 300 \text{ K} < \text{bulk PbTe}$$

ALL-SCALE HIERARCHICAL ARCHITECTURES

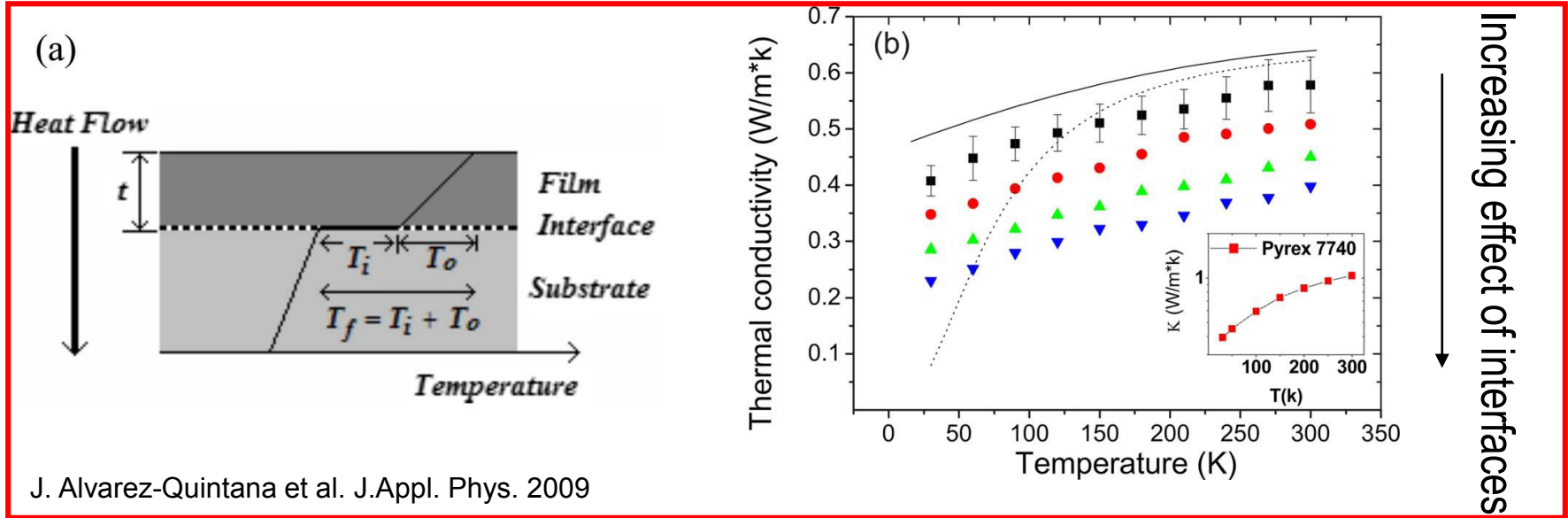
Na-doped PbTe:SrTe at 900 K

(solid-solution point defects, nm precipitates and gb's)
broad spectrum of heat-carrying phonons,

p-type PbTe (doping with Na)
nanostructured with SrTe at a
concentration of 4 mole % and
mesostructured with powder
processing and spark plasma
sintering

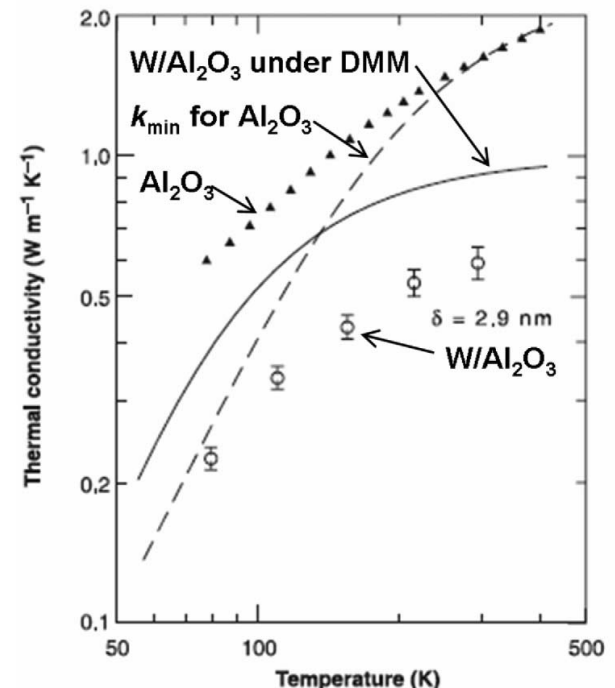


Below the amorphous limit: Taking benefit of interfaces



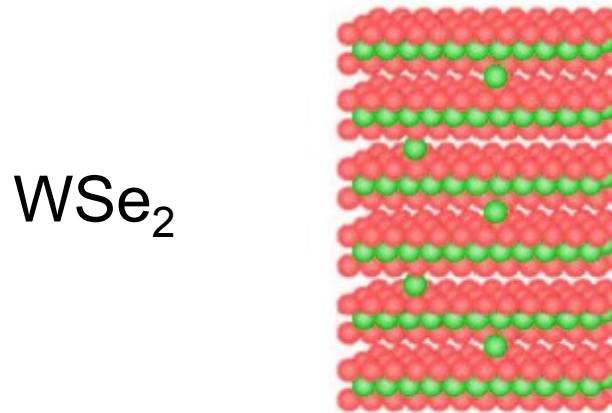
$$k_{eff} = \frac{k_{layers}}{1 + k_{layers} \cdot \frac{R_{th}}{t}}$$

W/Al₂O₃ nanolaminate with layers few nm thick
and
high density of interfaces has
 $\kappa < \text{amorphous limit}$



Ultralow k in VdW stacks

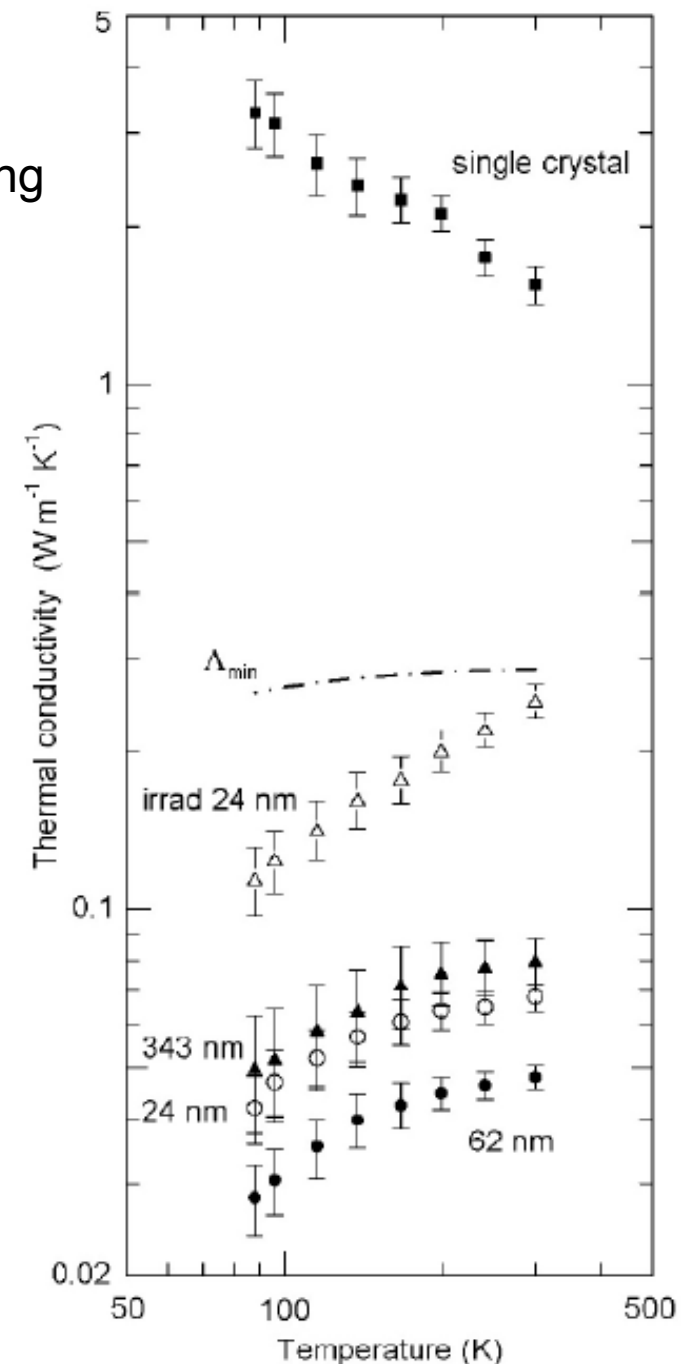
Sequential bilayers of W and Se by MBE + 1h annealing
for layered WSe_2 structure



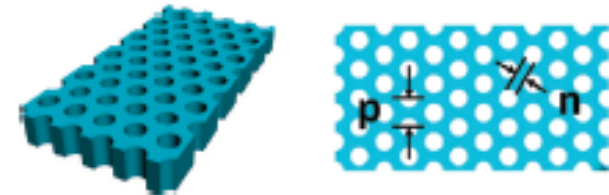
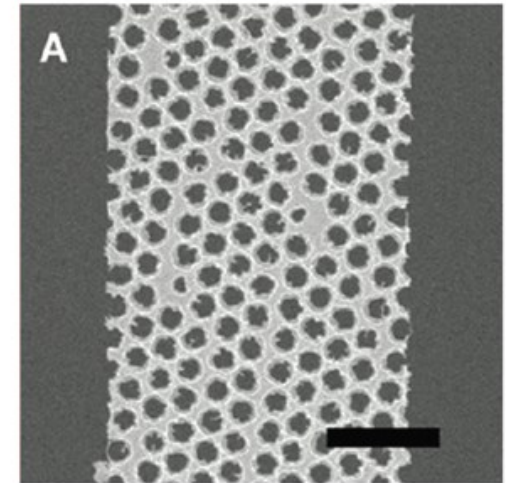
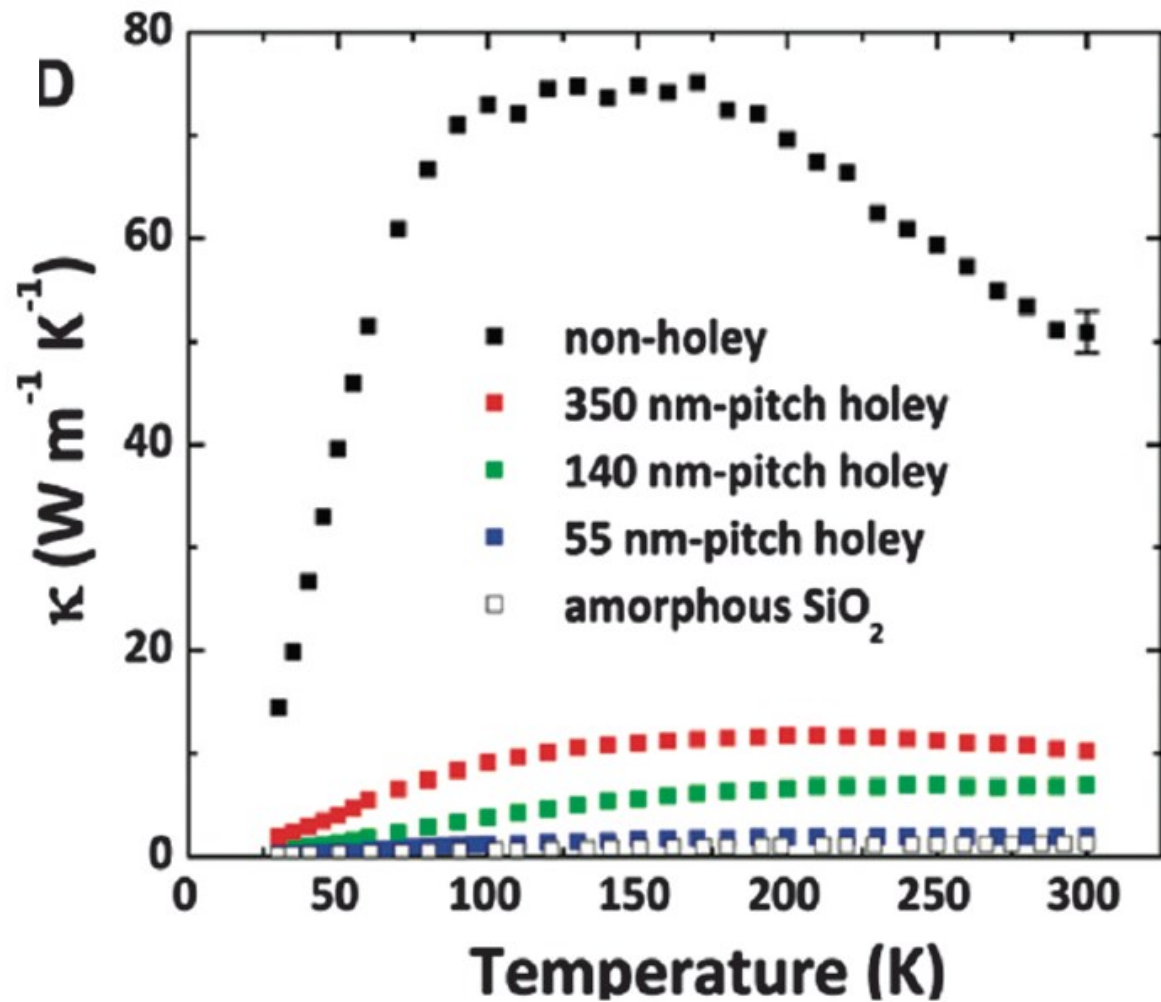
$$k(300\text{K}) = 0.048 \text{ W/mK}$$

6 times smaller than amorphous limit

VdW interaction between WSe_2 adjacent layers +
+ disorder in c-axis induces phonon localization

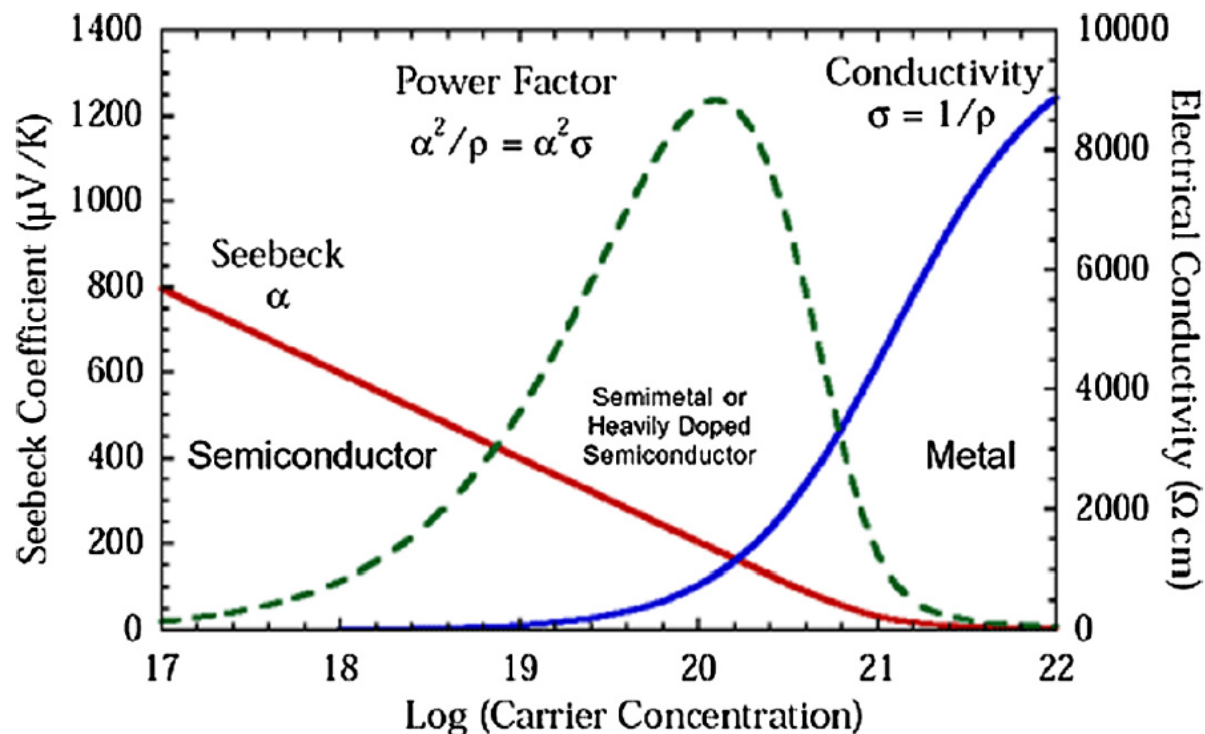


Nanostructured material: HOLEY SILICON



Strategies to increase power factor

$$S^2 \sigma$$



Mott relation

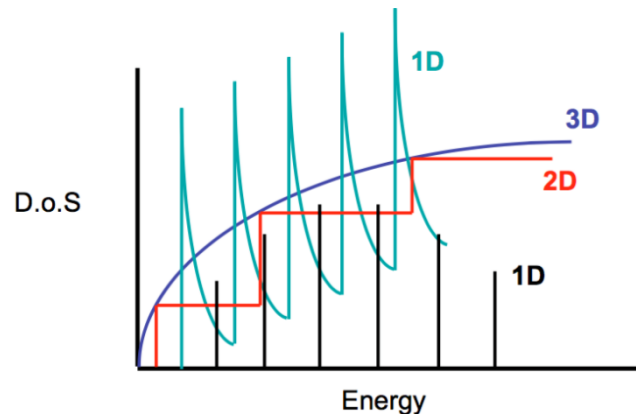
$$S = \frac{\pi^2}{3} \cdot \frac{k_B^2 T}{e} \cdot \left\{ \frac{\partial \log \sigma(\varepsilon)}{\partial E} \right\}_{E_F}$$

Any mechanism enhancing $(d\sigma/dE)$ will enhance S

(i) increasing the energy dependence of the conductivity close to the Fermi level

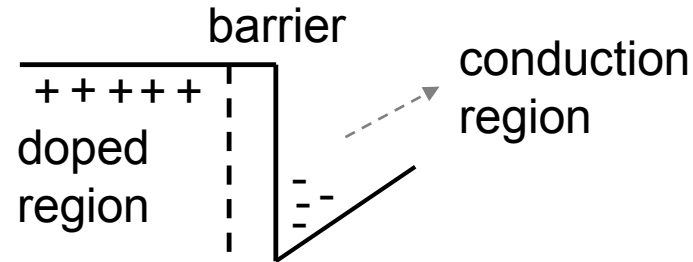
As $\sigma(E) = n(E) \cdot e \cdot \mu(E)$  Variations of $n(E)$ and mobility $\mu(E)$

(ii) increasing the energy dependence of $n(E)$ can be done by enhancing the energy dependence of DOS on energy, i.e. lowering the dimensionality



Modulated doping

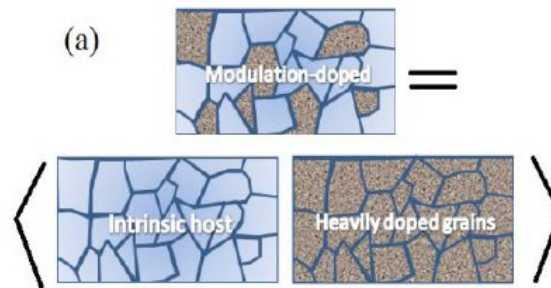
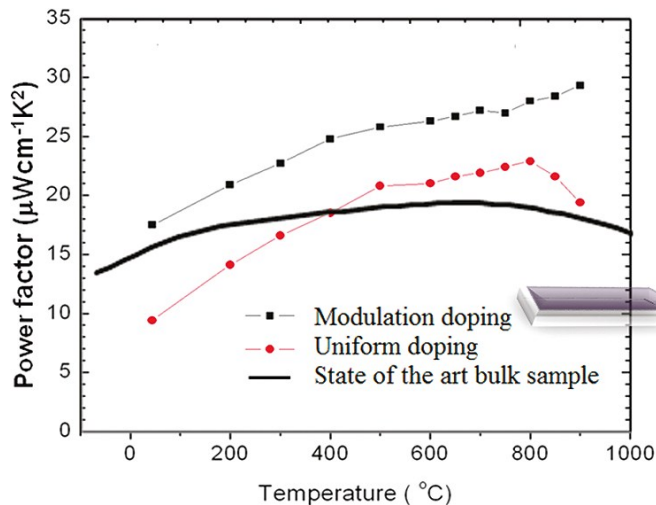
charge carriers are spatially separated from their parent impurity atoms to reduce the influence of impurity scattering increasing the electrical conductivity.



Mixing two types of SC NPs (20 nm):

undoped $(\text{Si}_{80}\text{Ge}_{20})_{70}$ - doped $(\text{Si}_{100}\text{B}_5)_{30}$ or doped $(\text{Si}_{100}\text{P}_3)_{30}$

+ Compaction two types of nanoparticles to form a bulk composite.



doping

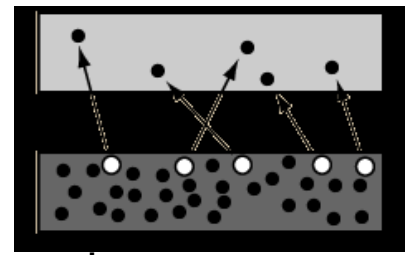
$$2.6 \times 10^{20} \text{ cm}^{-3}$$

Mobility

30% > in modulated sample

Energy Filtering (I)

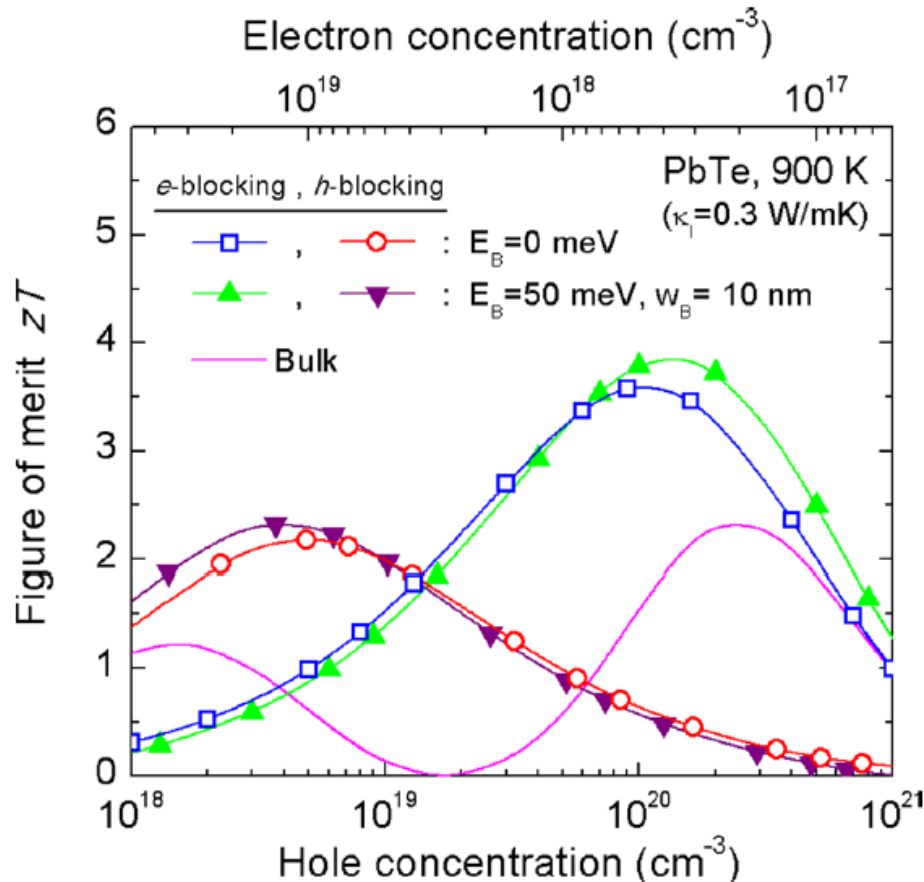
Minority carrier blocking



As T increases bipolar k increases lowering S

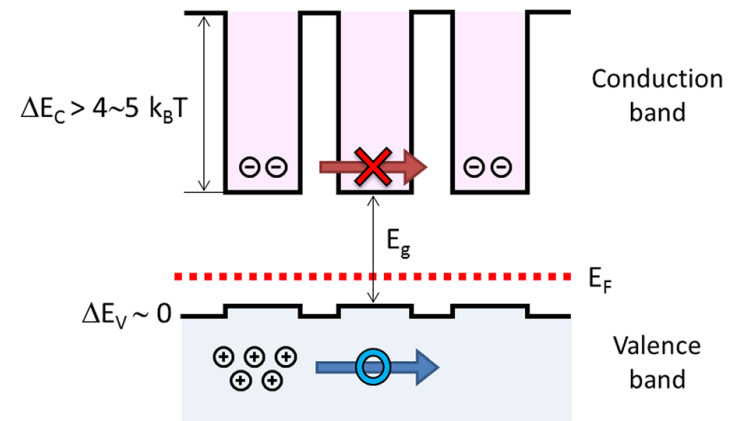
In the intrinsic regime

$$\kappa_{bi} \propto \exp\left(-\frac{E_g}{2k_B T}\right)$$



Concept

Blocking minority carriers

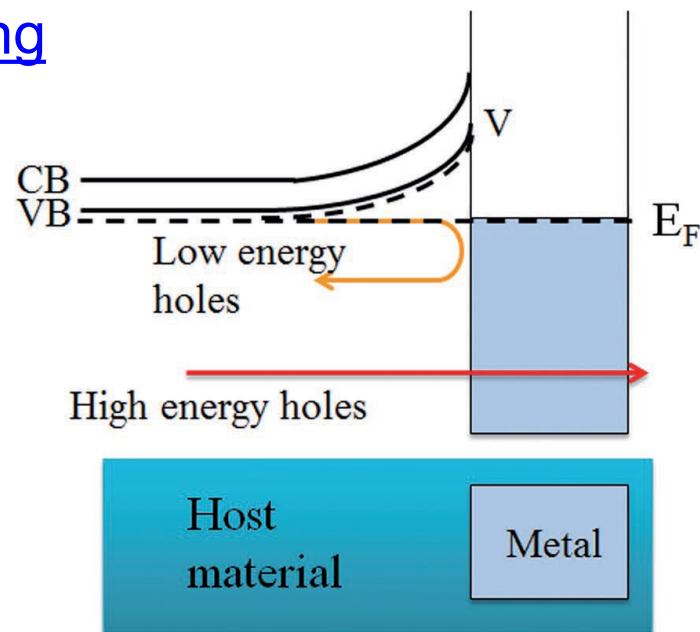


SLs or NP embedded in matrix

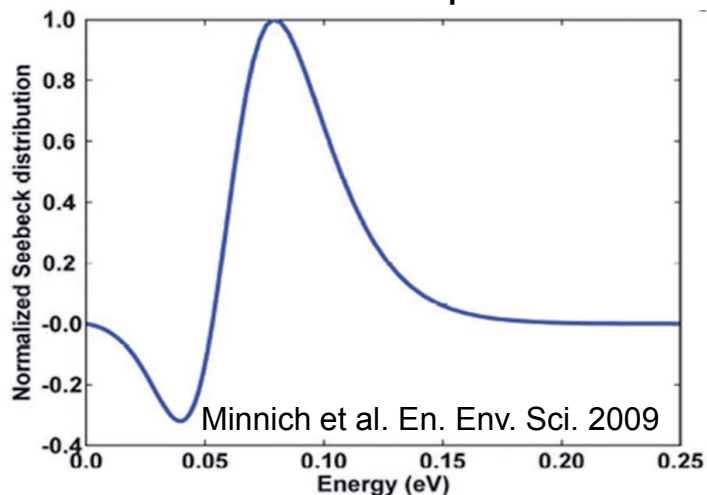
p-type PbTe:SrTe system

Energy Filtering (II) [Hot carrier filtering](#)

Second-phase NPs or specific barriers (gb & others) can serve as potential barriers that prohibit low energy carriers from passing through the barriers



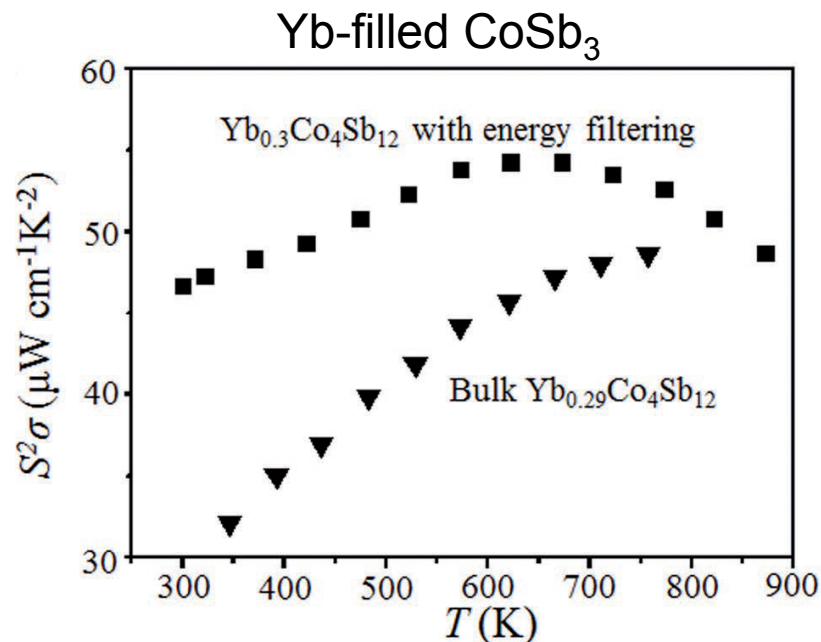
Calculation for p-SiGe



By filtering low-energy carriers average carrier energy increased

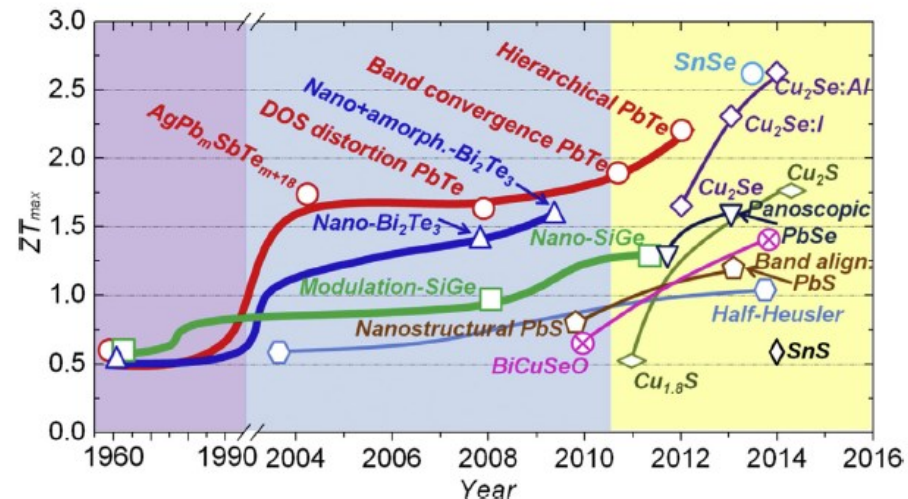
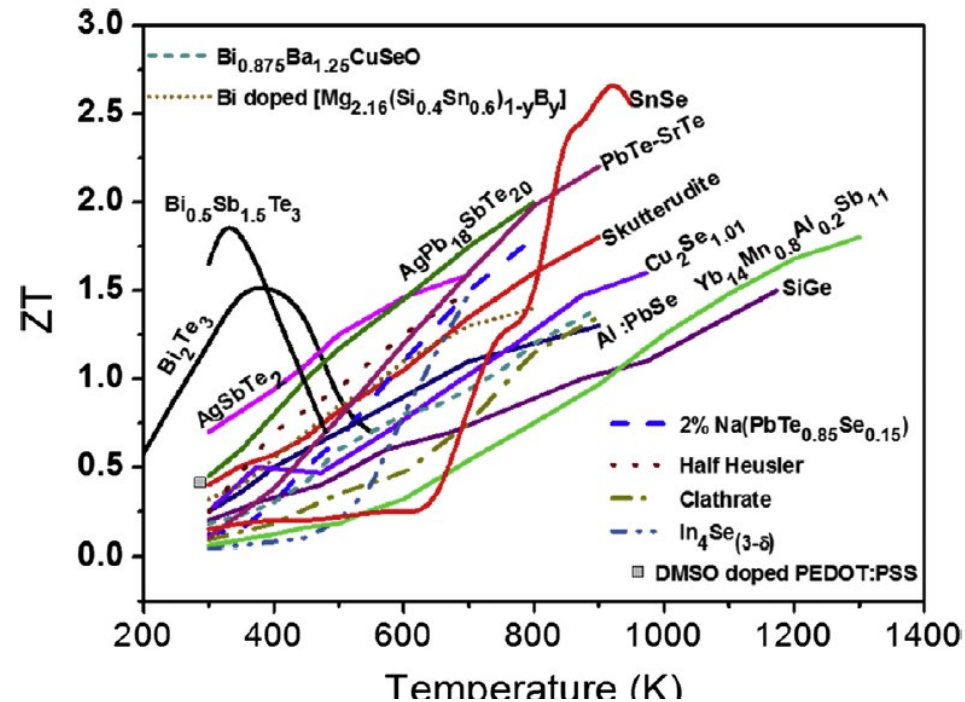
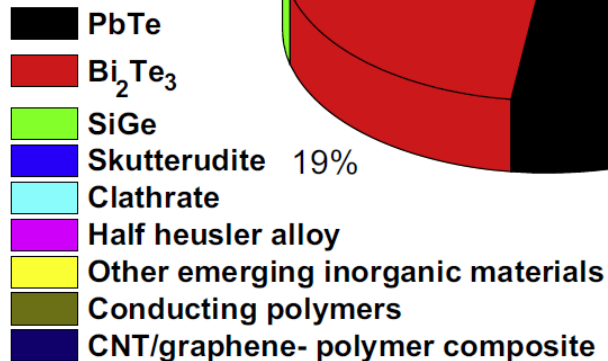
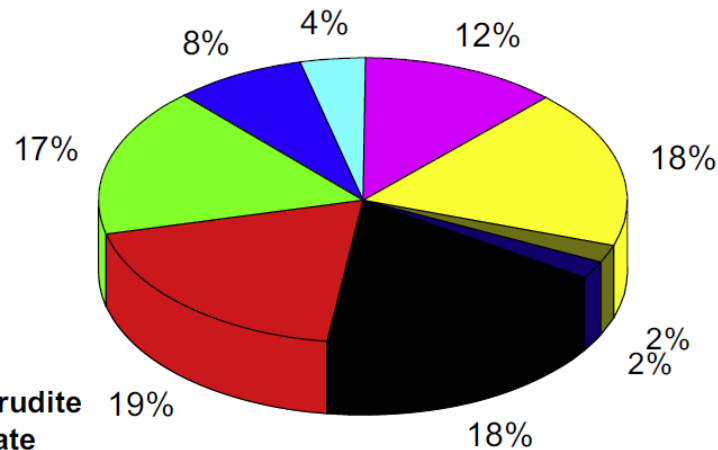


increase Seebeck coefficient



TE MATERIALS

- Bulk tellurides
- SiGe
- Half-Heuslers
- Skutterudites
- Zintl phases, group 14 clathrates
- Metal oxides
- Nanostructured materials
 - Superlattices

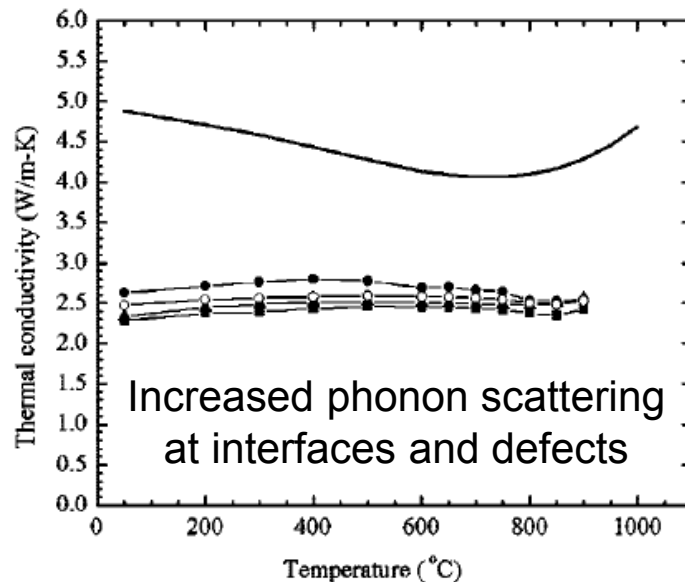


Nanostructured materials. Case of SiGe

SiGe alloys: traditional material for high T Applications ($T > 800^\circ\text{C}$)

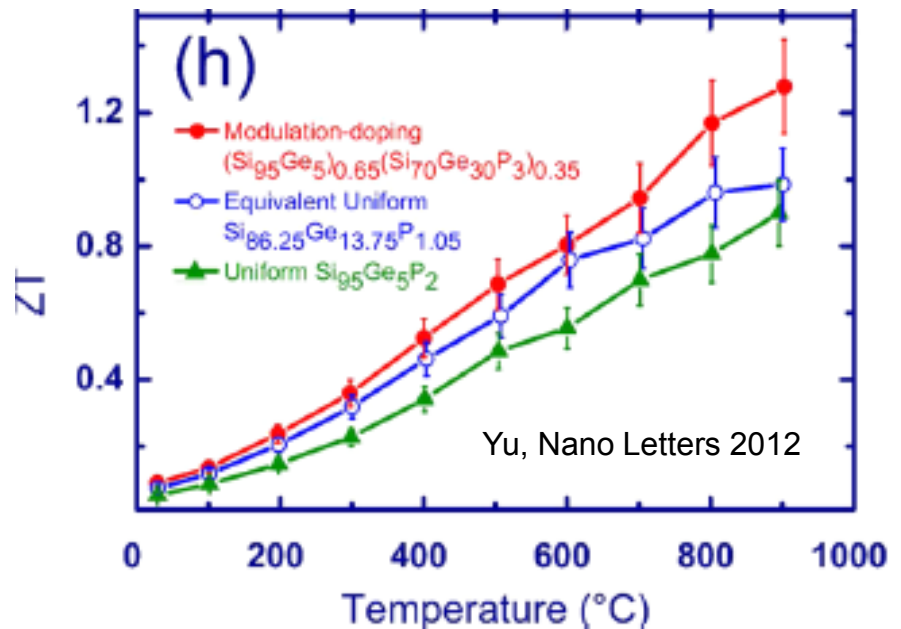
Can be doped n- and p-type enabling complete TE devices in one platform

Fabrication: BM + HP



n-type ZT up to 1.5 at $T = 1173\text{ K}$
p-type ZT up to 0.95

$\text{Si}_{80}\text{Ge}_{20}$ with grain sizes about 20 nm



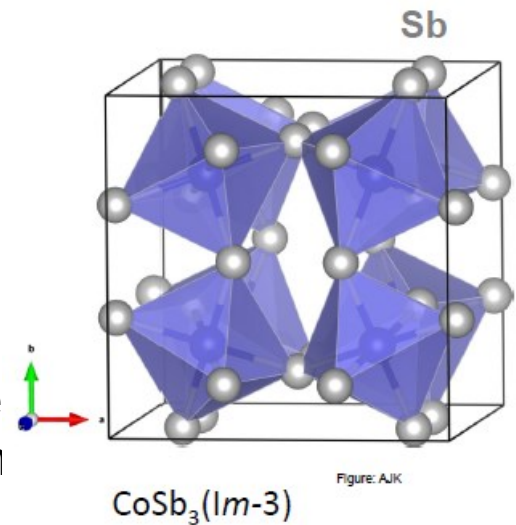
Increase ZT is a general feature
in nanograin materials

Skutterudites

“skutterudite” refers to naturally occurring mineral CoAs_3 , first discovered in Skutterud, Norway.

Cubic structure composed of 32 atoms per unit cell

CoAs_3 exhibits a distorted version of the AB_3 -type perovskite structure, that is, MX_3 (M = Co, Rh, Ir is a group 9 transition metal and X a group 15 nonmetal P, As, Sb), with an octahedral structure and a void at its center.



Key aspects:

Filling the void fulfills PGEC-concept such as in $\text{CeFe}_4\text{Sb}_{12}$

The void-filling atom can act as an electron donor or electron acceptor.
Strong phonon scattering centers reduce k .

Good thermal and mechan. properties of both n- and p-type operating at medium T

Material		n/p	ZT_{MAX}	Typical operating T (°C)	Max operating T (°C)
CoSb_3	PF=6.5 $\mu\text{W}/\text{cmK}^{-2}$, $k=10$ W/mK	n	0.8	500–600	700
$\text{CeFe}_4\text{Sb}_{12}$		p	0.8	500–600	700

Skutterudites

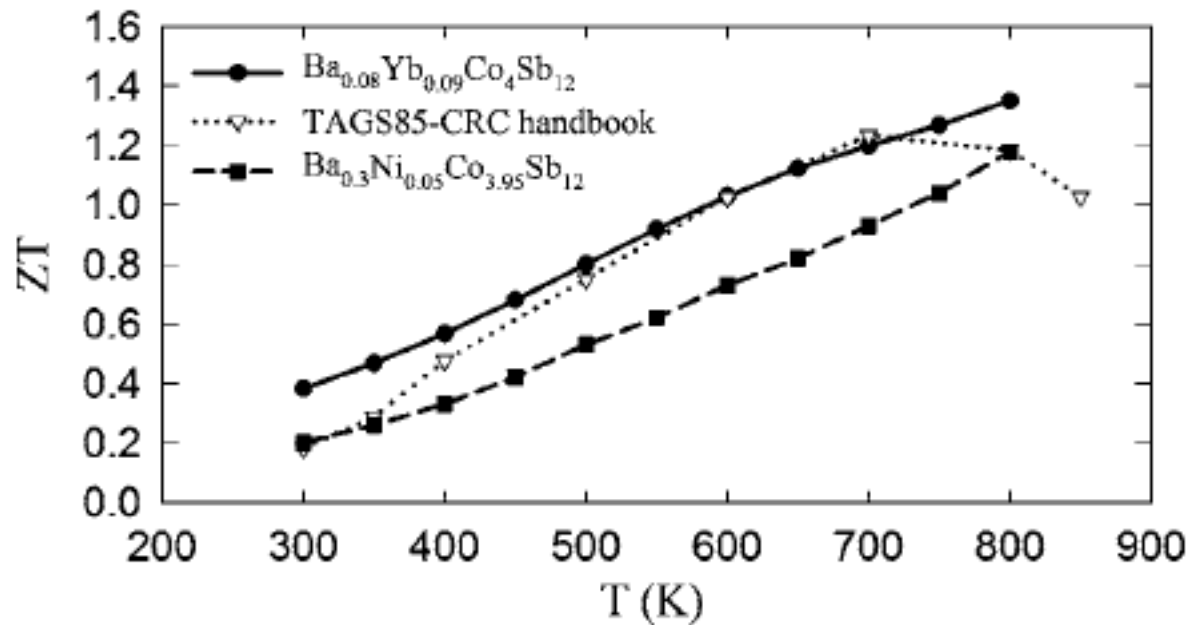
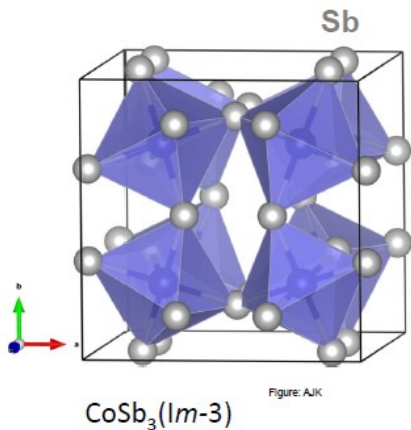
Multiple filling of voids

A ZT value of 1.36 is obtained for n-type $\text{Ba}_x\text{Yb}_y\text{Co}_4\text{Sb}_{12}$ at 527 C.

The multiple-filled skutterudite $\text{Ba}_{0.08}\text{La}_{0.05}\text{Yb}_{0.04}\text{Co}_4\text{Sb}_{12}$ exhibits a ZT of 1.7 at 577°C.

Filler ions form phonon resonant scattering centers with its own frequency.

Multiple choice of filling atoms facilitates that different normal mode phonons of the lattice can scatter with the filler decreasing κ



Half-Heusler alloys (potential for high T)

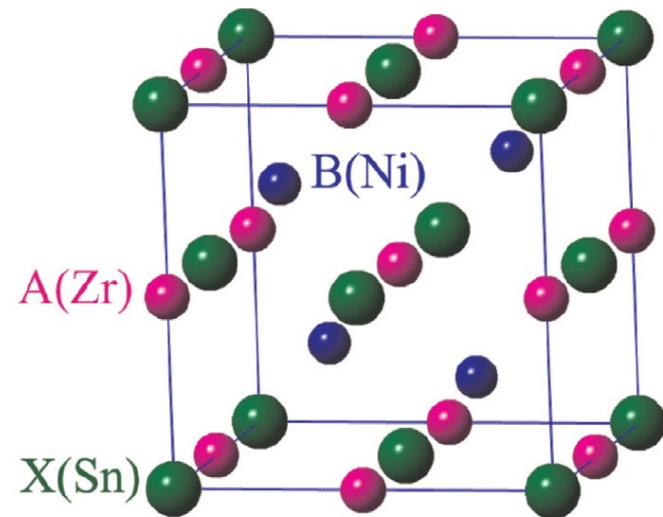
Intermetallic phases with formula ABX formed by 3 interpenetrated fcc sublattices and one vacant sublattice

MCoSb and MNiSn with **M**=Ti, Zr, Hf

high S , high σ , high k

Substitution of A,B sites to reduce k due to mass fluctuations and strain fields

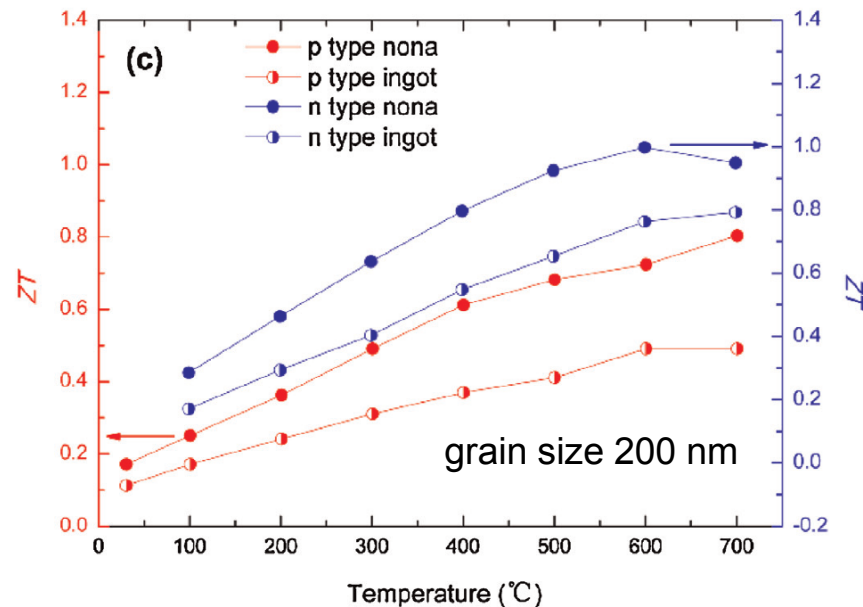
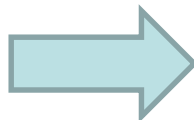
Substitution of X sites to tune carrier density and therefore increase σ



Nanocomposite approach in HH

k_{lat} decreases 30% for
p-type $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{CoSb}_{0.8}\text{Sn}_{0.2}$

k_{lat} decreases 25% for
n-type $\text{Hf}_{0.75}\text{Zr}_{0.25}\text{NiSn}_{0.99}\text{Sb}_{0.01}$



Hf and Ti reduces k compared to Hf and Zr

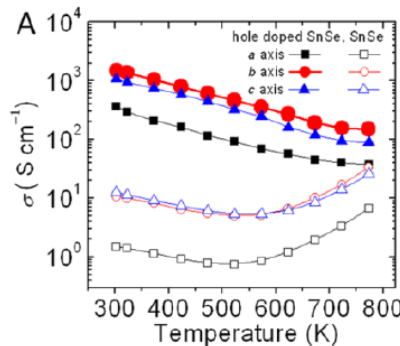
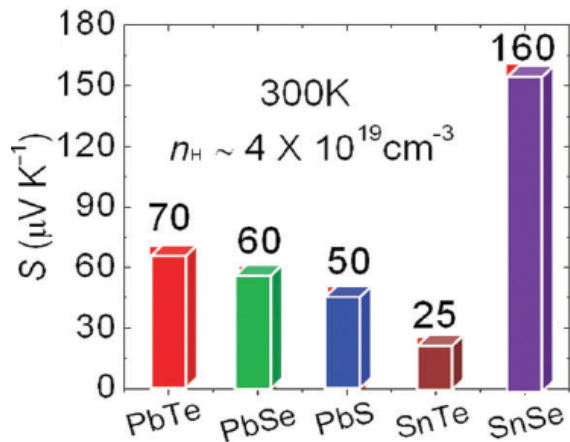
Single-Crystal SnSe

ZT=2.6 at 973 K

SnSe does not have high MW,
nanostructuring,
a complex crystal structure
or a large unit cell.

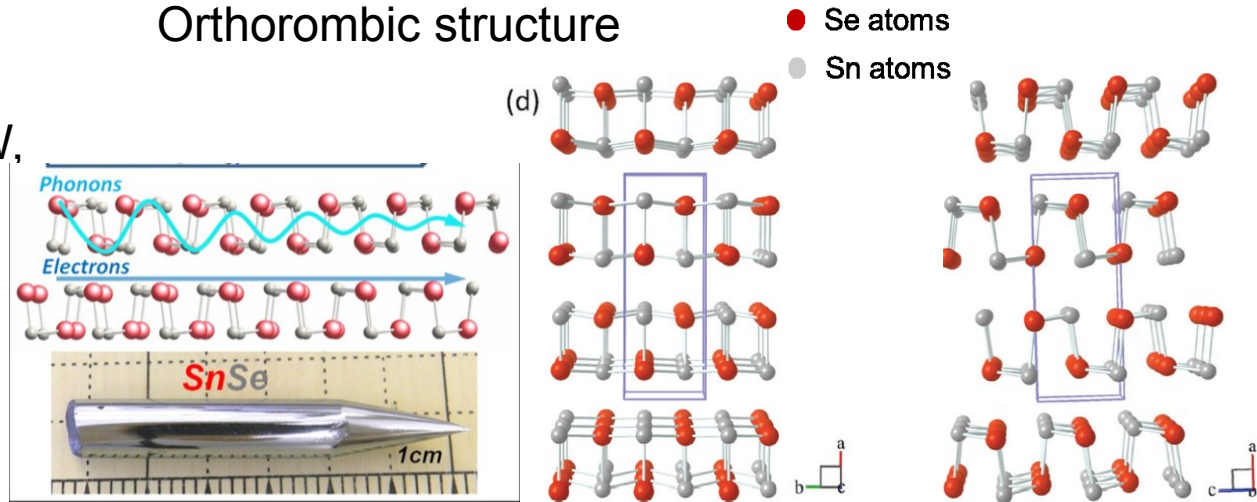
Power factor increase
to $40 \mu\text{W cm}^{-1} \text{K}^{-2}$

$$S = \frac{\pi^2}{3} \cdot \frac{k_B^2 T}{e} \cdot \left\{ \frac{\partial \log \sigma(\epsilon)}{\partial E} \right\}_{E_F}$$

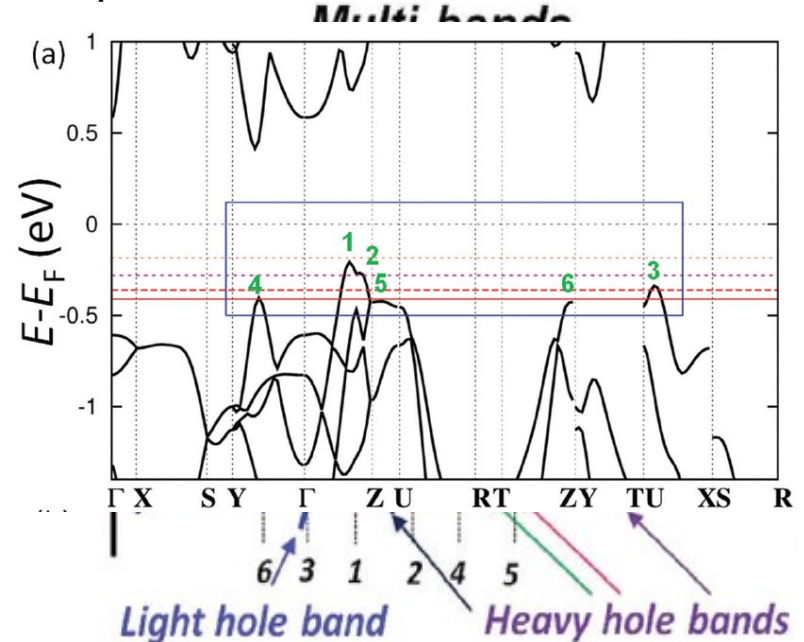


Simultaneous low k and higher power factor

Orthorhombic structure

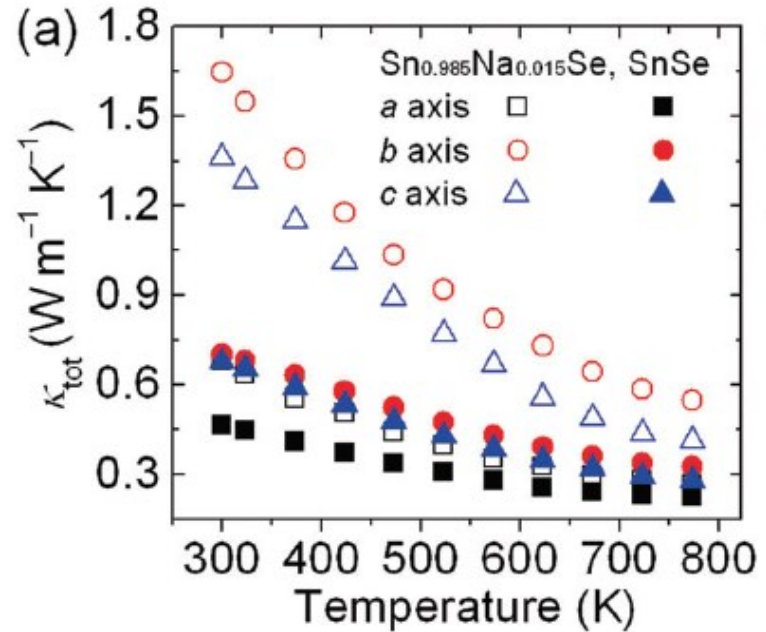
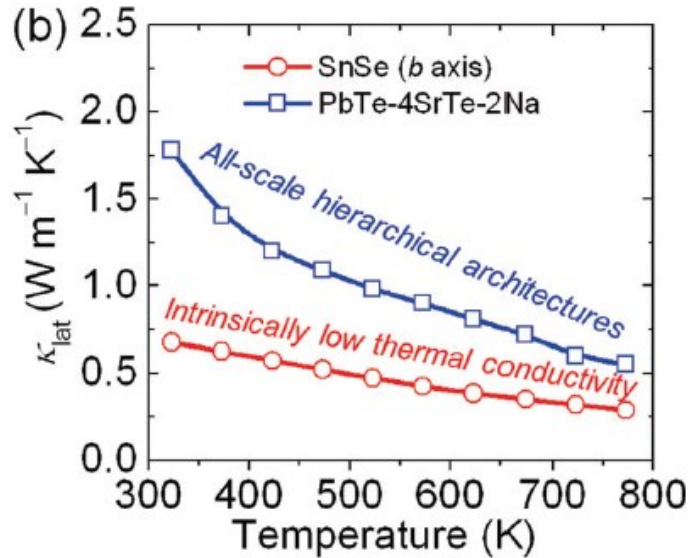


Complex electronic structure



Single-Crystal SnSe

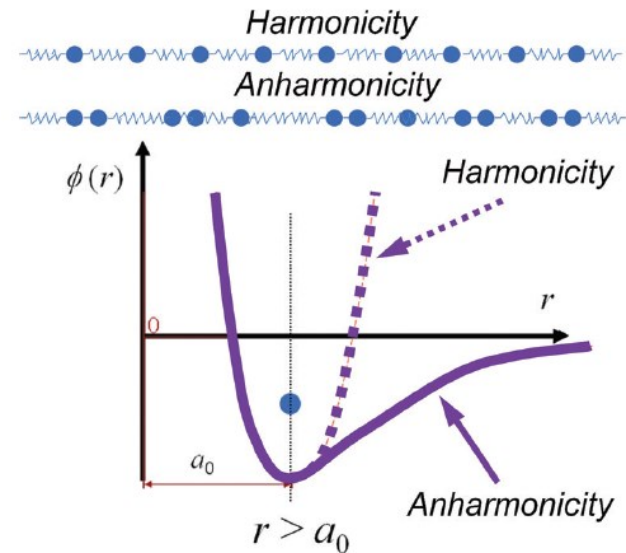
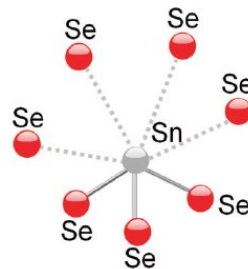
Low lattice thermal conductivity (SnSe)



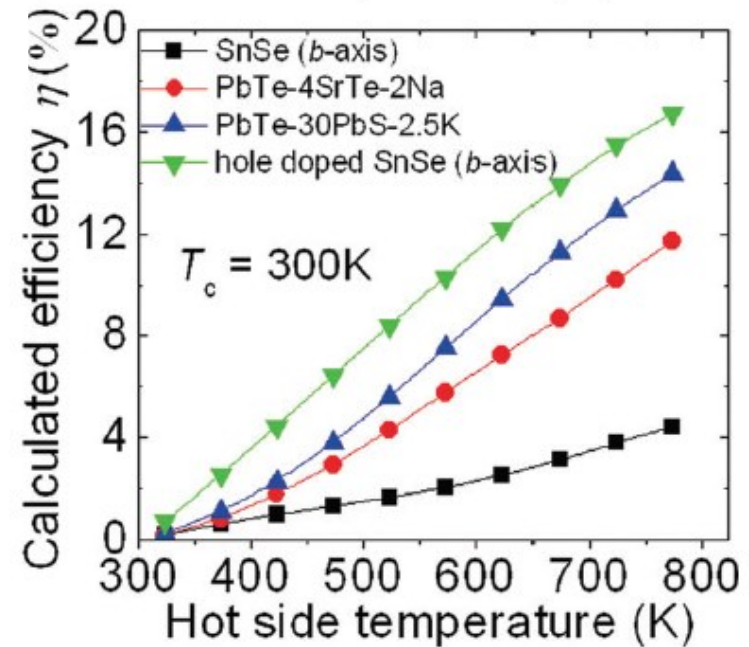
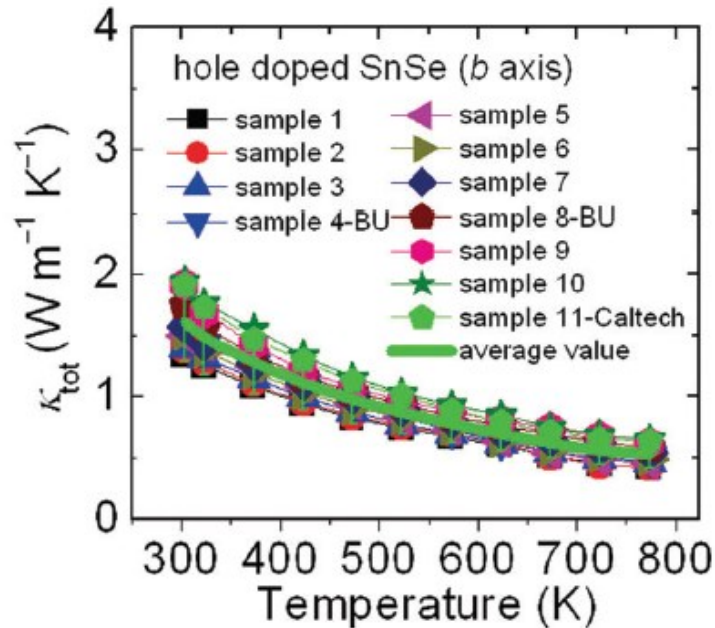
Low value of k partly comes from the 2D nature of the material but other effects should be invoked

High anharmonicity results in strong ph-ph scattering reducing k

Heavily distorted SnSe₇ polyhedra



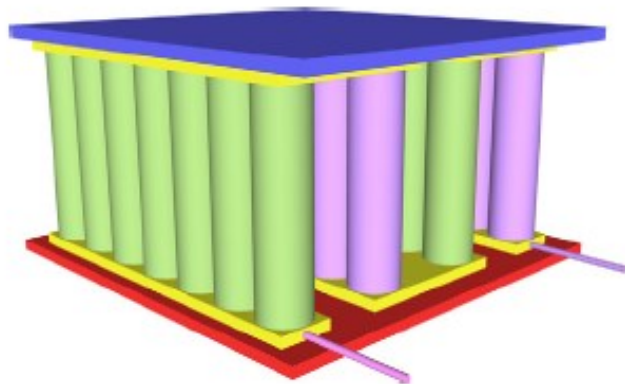
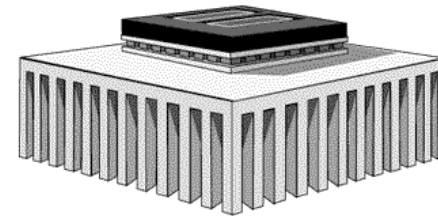
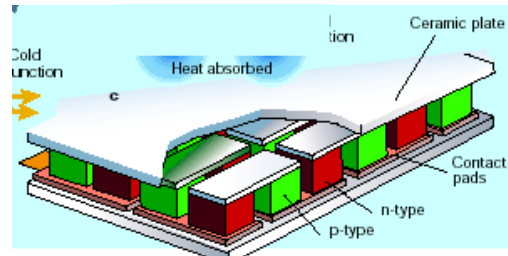
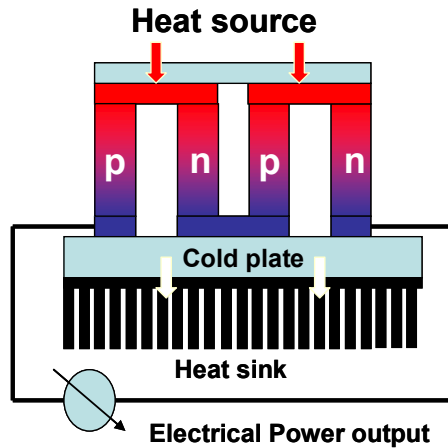
Single-Crystal SnSe



The physics of thermal and charge transport in SnSe is unusual and fascinating. The high thermoelectric performance of SnSe crystals suggests that single phase materials with strongly anharmonic bonding and ultralow thermal conductivity are promising candidates for achieving high thermoelectric performance.

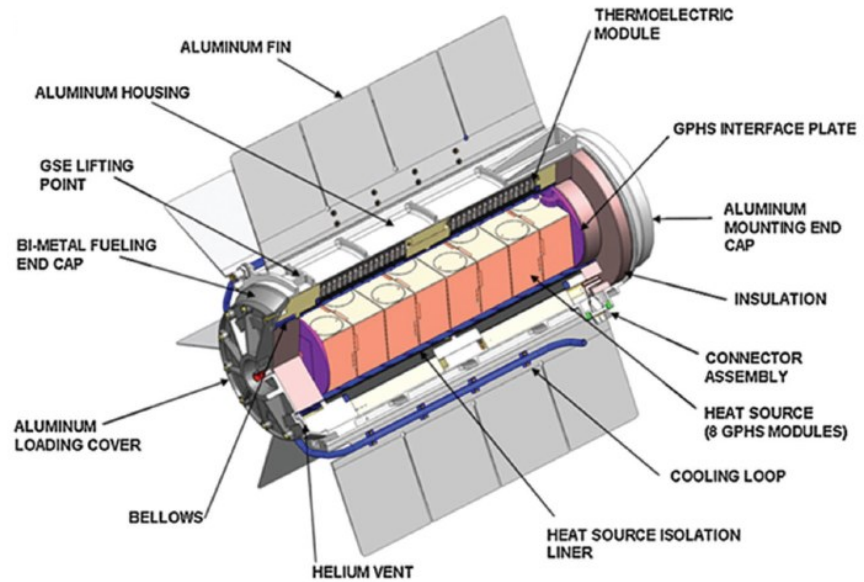
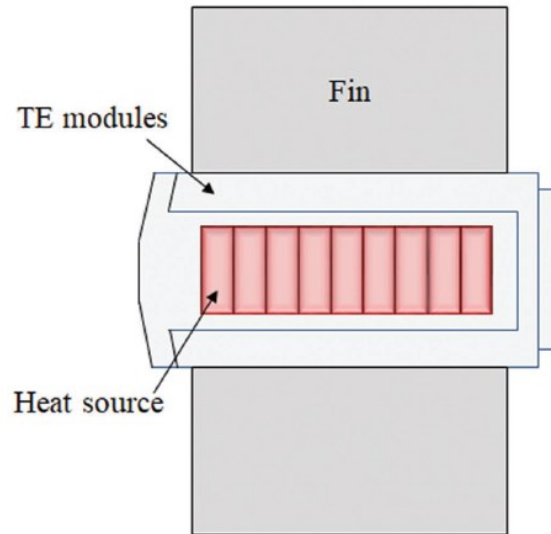
APPLICATIONS

TE generators: Set of n,p legs electrically connected in series and thermally in parallel



Apart from thermoelectric materials, contact materials also play a key role in TE devices. The semiconductor–metal junction exhibits a greater electrical resistance than a metal–metal junction, which affects the performance of the device

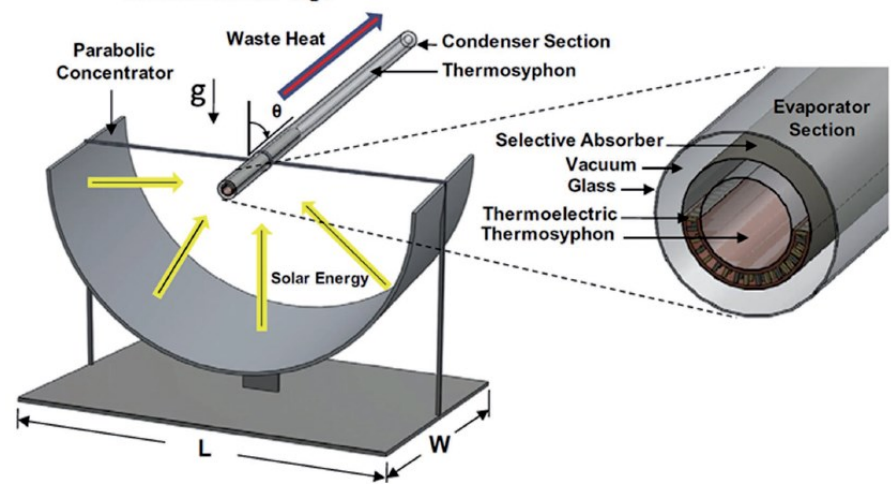
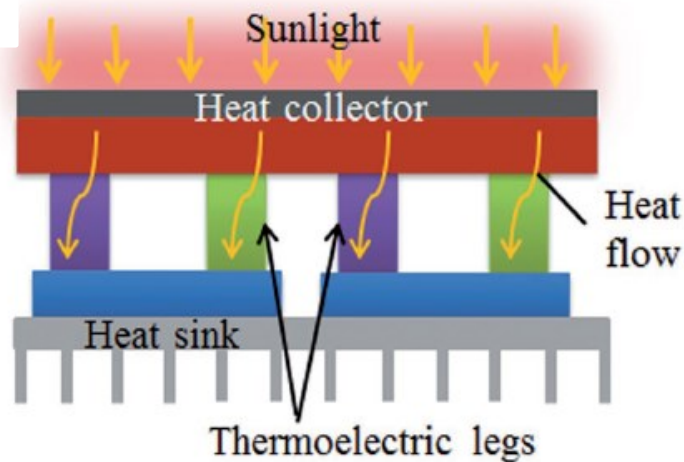
Radioisotope thermoelectric generators, RTGs



Yang, Adv. Energy Mat. 2017

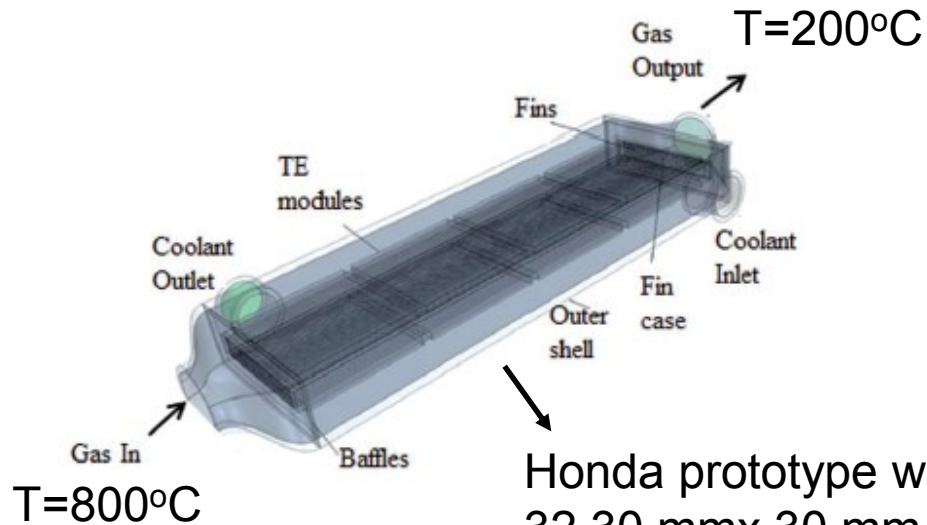
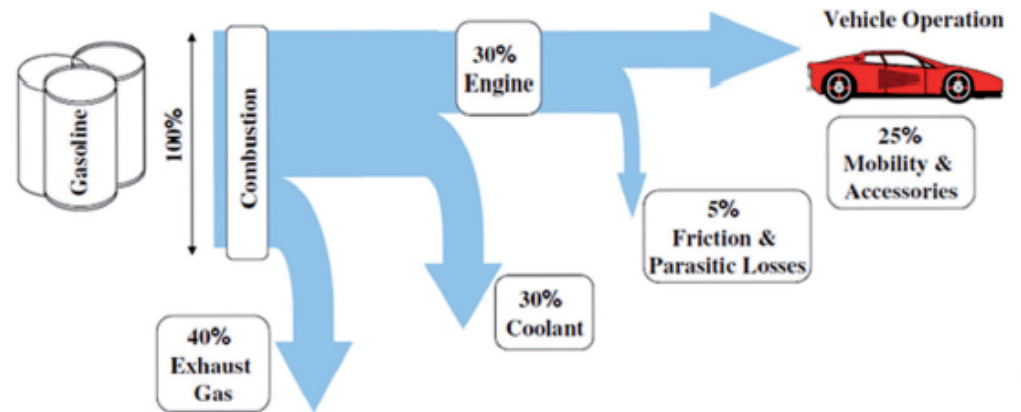
ZT of 0.5 (p-type) and 0.9 (n-type)

Solar Thermal Power Generation (STEG's)



Materials		ΔT [K]	N^a	$P_{\max}^b$ [W]	$\eta_{\max}$ [%]
p-type	n-type				
Bi-Te alloy	Bi-Te alloy	92	6 modules	4.7	1.2
(Bi ₂ Te ₃) ₁ (CdTe) core-shell nanostructured thin film				16.66 mA cm ⁻²	4.8
Bi _{0.5} Sb _{1.5} Te ₃ thin film	Bi ₂ Te _{2.7} Se _{0.3} thin film	100		80×10^{-6}	
Commercial GM-200-127-14-16		≈10	127	4.7×10^{-3}	0.23
TGM-127-1.4-2.5		155		3	4

AUTOMOTIVE TEG's



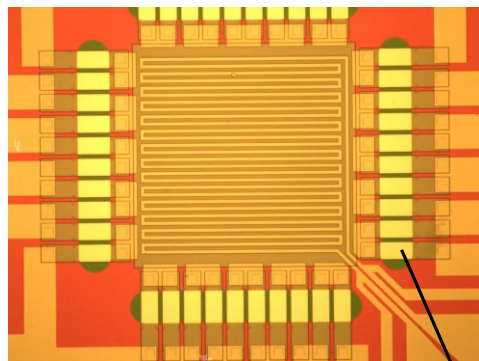
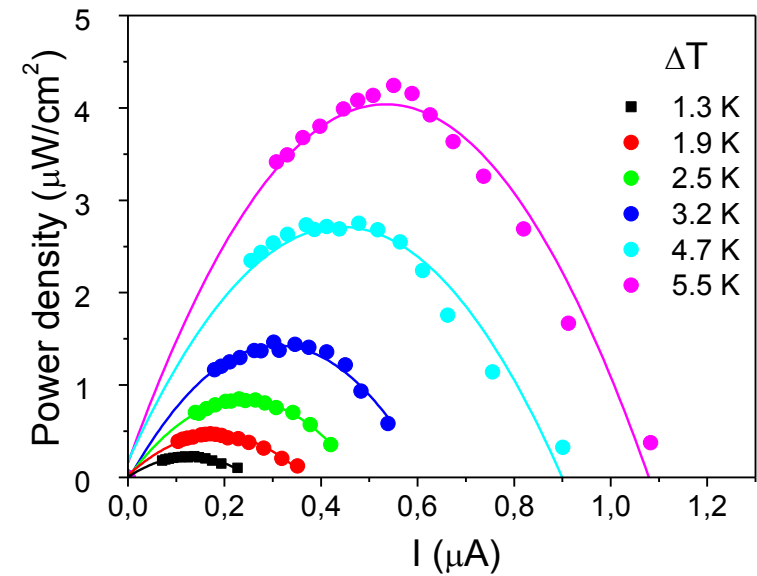
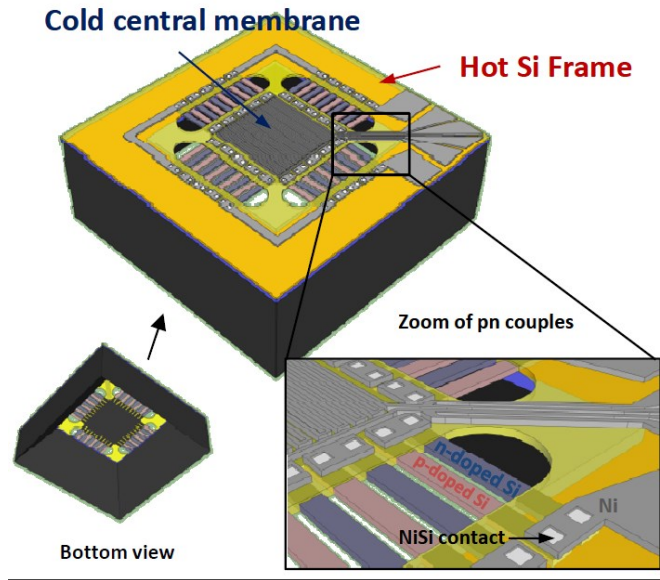
Still in concept stages

Goal total of 750 W from 20 W TEGs

400 W with 4.6 kg of TE material

Honda prototype with liquid cooler gives 500 W
32 30 mmx 30 mm TEGs

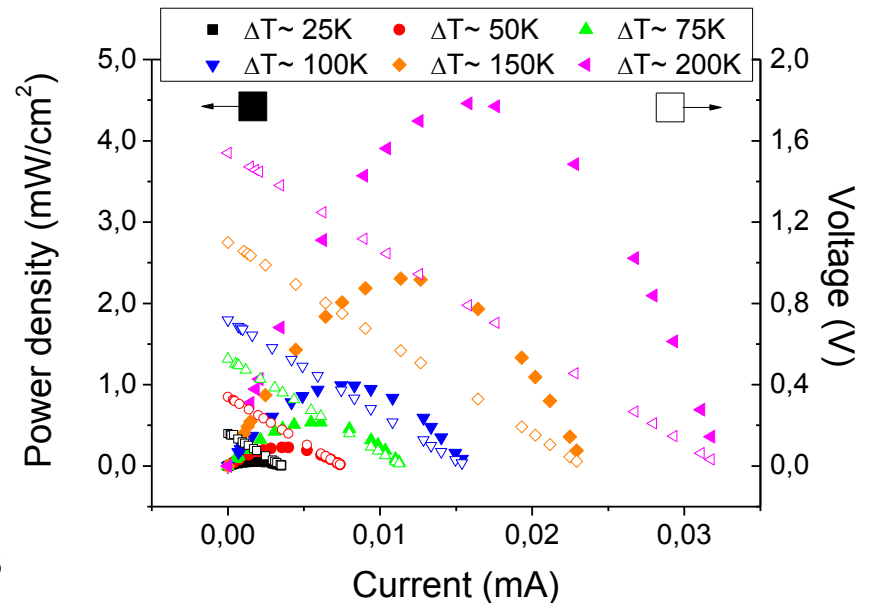
Microthermoelectric TE-based generators/sensors



UAB, Nano Energy 2015

Size: $3 \times 3 \text{ mm}^2$

Low-d Si thermopiles



Low Power: using body heat

Thermoelectric watch



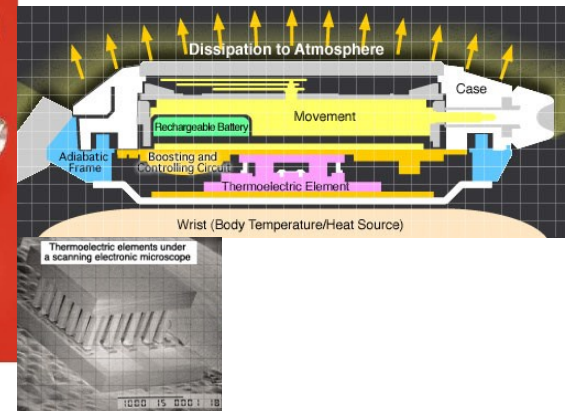
Bulova Thermatron (1982). Unreliable technology. The company went bankrupt.

<http://www.watchonista.com/2914/watchonista-blog/news/bulova-thermatron-%E2%80%93-thermal-revolution>



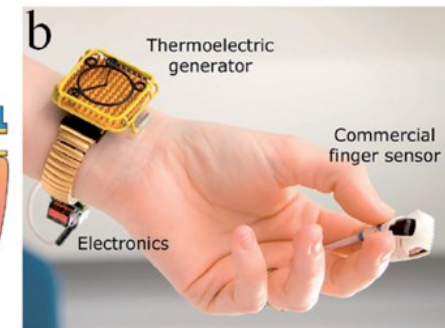
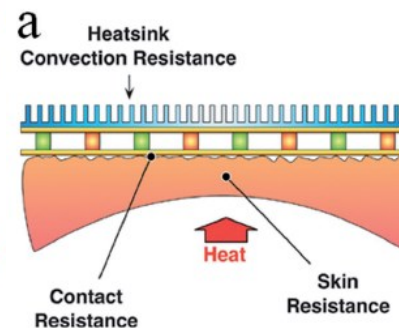
Seiko Thermic

Seiko Thermic (1998). This one actually worked. The watch still needs a battery.



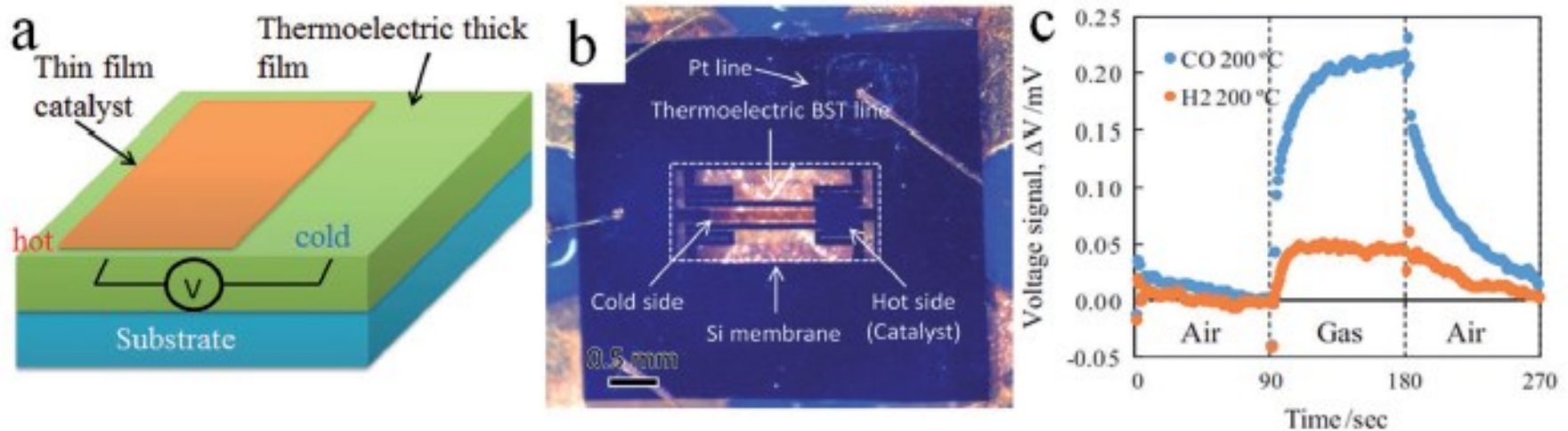
TE watches refuse to die: MATRIX PowerWatch crowdfunded at Indiegogo (2016)

<https://www.indiegogo.com/projects/smartwatch-powered-by-you-matrix-powerwatch-watch-fitness>



SENSING APPLICATIONS

Detecting heat signals from catalytic reactions

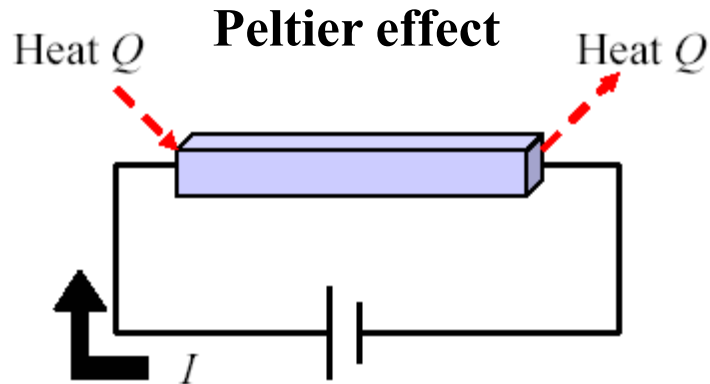


Pt thin film acts as a catalyst for the reaction with H₂, releasing heat and creating a T difference

Good sensitivity and short response time

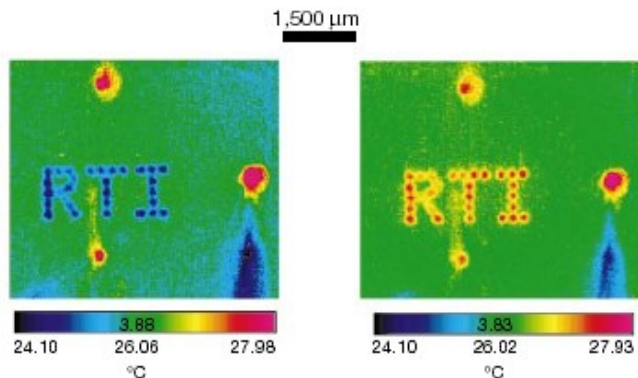
Other catalyst can be used for other gases

On-chip cooling using SL's

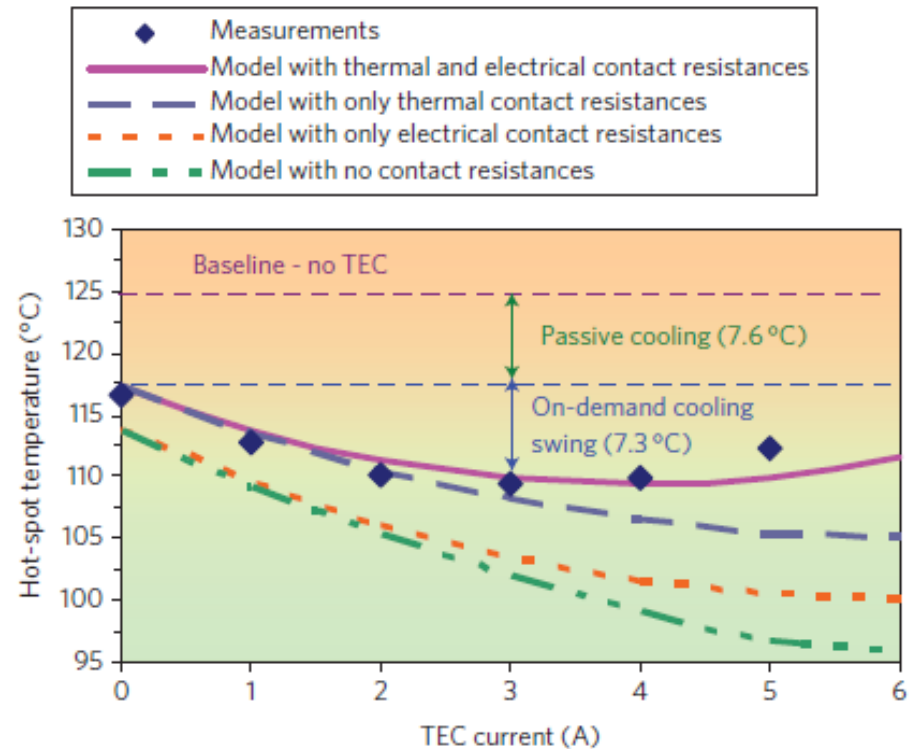
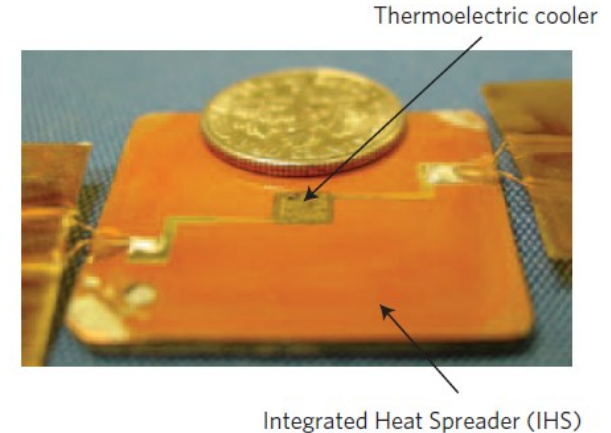


Creation of a heat difference from an electric voltage

$$\Pi = Q/I \quad \text{Heat carried by carriers}$$



Venkatasubramanian et al. Nature, (2001) 597-.



Choudhury et al. Nat. Nanotech 2007

Thermoelectric refrigerators

- Electric current generates a temperature gradient
 - Can be used for cooling (or heating!)
- Thin-film Peltier coolers do actually possess high cooling power densities
- They can be utilized in scenarios where compressor-based cooling is impossible



Figure: blog.novaeletronica.com.br

Figure: alphageek.fi



USB-powered Peltier refrigerator

CONCLUSIONS

- ❑ Reducing k below the alloy limit may require new strategies such as reduction of the phonon group velocity or reducing the number of phonon modes that propagate.
Anharmonicity may play a significant role for decreasing k .
- ❑ Many of strategies to increase ZT. However, ZT above 3 in both n- and p-type materials (required for some dreamed applications of energy harvesting) still far to be realized.

Still PLENTY of ROOM for IMPROVEMENT

- ❑ In spite of this there are many opportunities for low-power applications to power sensor or low-consumption devices.

THANK YOU